VORON



VORON STEALTHBURNER

We build space shuttles with gardening tools
so anyone can have a space shuttle of their own.

VERSION 2026-11-07



Before you begin on your journey, a word of caution:

You are about to assemble a robot in the comfort of your own home. This machine can maim, burn, and electrocute you if you are not careful. Please do not become the first VORON fatality. There is no special Reddit flair for that.

Please read the entire manual before you start assembly. As you begin wrenching, please check our Discord channels for any tips and questions that may halt your progress.

Most of all, good luck!

THE VORON TEAM

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PART PRINTING GUIDELINES

The VORON Team has provided the following print guidelines for you to follow in order to have the best chance at success with your parts. There are often questions about substituting materials or changing printing standards, but we recommend that you follow these:

3D PRINTING PROCESS

Fused Deposition Modeling (FDM)

MATERIAL

ABS or ASA

LAYER HEIGHT

Recommended: 0.2mm

EXTRUSION WIDTH

Recommended: Forced 0.4mm

INFILL TYPE

Grid, Gyroid, Honeycomb, Triangle or Cubic

INFILL PERCENTAGE

Recommended: 40%

WALL COUNT

Recommended: 4

SOLID TOP/BOTTOM LAYERS

Recommended: 5

PRINT IT FORWARD (PIF)

Often times community members that have issues printing ABS will bootstrap themselves into a VORON using our Print It Forward program. This is a service where approved members with VORON printers can make you a functional set of parts to get your own machine up and running. Check Discord if you have any interest in having someone help you out.

FILE NAMING

By this time you should have already downloaded our STL files from the VORON GitHub. You might have noticed that we have used a unique naming convention for the files. This is how to use them.

PRIMARY COLOR

Example z_joint_lower_x4.stl

These files will have nothing at the start of the filename.

ACCENT COLOR

Example [a]_tensioner_left.stl

We have added “[a]” to the front of any STL file that is intended to be printed with accent color.

QUANTITY REQUIRED

Example: [a]_z_belt_clip_lower_x4.stl

If any file ends with “_x#”, that is telling you the quantity of that part required to build the machine.

HOW TO GET HELP

If you need assistance with your build, we’re here to help. Head on over to our Discord group and post your questions. This is our primary medium to help VORON Users and we have a great community that can help you out if you get stuck.

DISCORD

https://discord.gg/voron

VORON FORUM

https://forum.vorondesign.com

REPORTING ISSUES

Should you find an issue in the documentation or have a suggestion for an improvement please consider opening an issue on GitHub (https://github.com/VoronDesign/Voron-Stealthburner/issues). When raising an issue please include the relevant page numbers and a short description; annotated screenshots are also very welcome. We periodically update the manual based on the feedback we get.

THIS IS JUST A REFERENCE

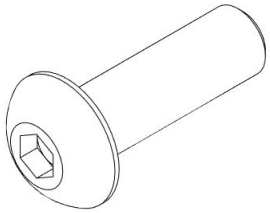
This manual is designed to be a simple reference manual. Building a VORON can be a complex endeavour and for that reason we recommend downloading the CAD files off our Github repository if there are sections you need clarification on. It can sometimes be easier to follow along when you have the whole assembly in front of you.



https://github.com/vorondesign



https://docs.vorondesign.com



BUTTON HEAD CAP SCREW

Metric fastener with a domed shape head and hex drive. Most commonly found in locations where M5 fasteners are used.

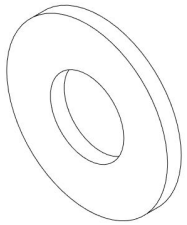
ISO 7380-1



FLAT HEAD CAP SCREW

Metric fastener with a flat head and hex drive.

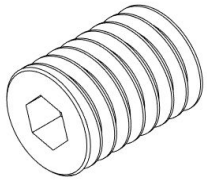
ISO 10642



WASHER

Usually stamped from sheet metal this type of spacer is not as consistent in thickness as the shims are. Only used in M3 size.

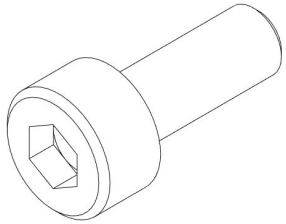
DIN 125



SET SCREW

Small headless screw with an internal drive. Used in pulleys and other gears. Also called a grub screw.

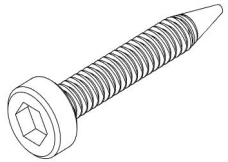
ISO 4026



SOCKET HEAD CAP SCREW

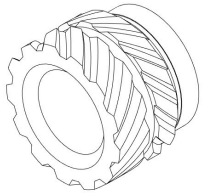
Metric fastener with a cylindrical head and hex drive. The most common fastener used on the Voron.

ISO 4762



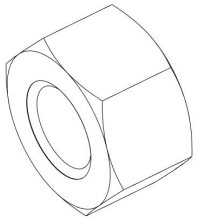
SELF TAPPING SCREW

Fastener with a pronounced thread profile that is screwed directly into plastic.



HEAT SET INSERT

Heat inserts with a soldering tip so that they melt the plastic when installed. As the plastic cools, it solidifies around the knurls and ridges on the insert for excellent resistance to both torque and pull-out.



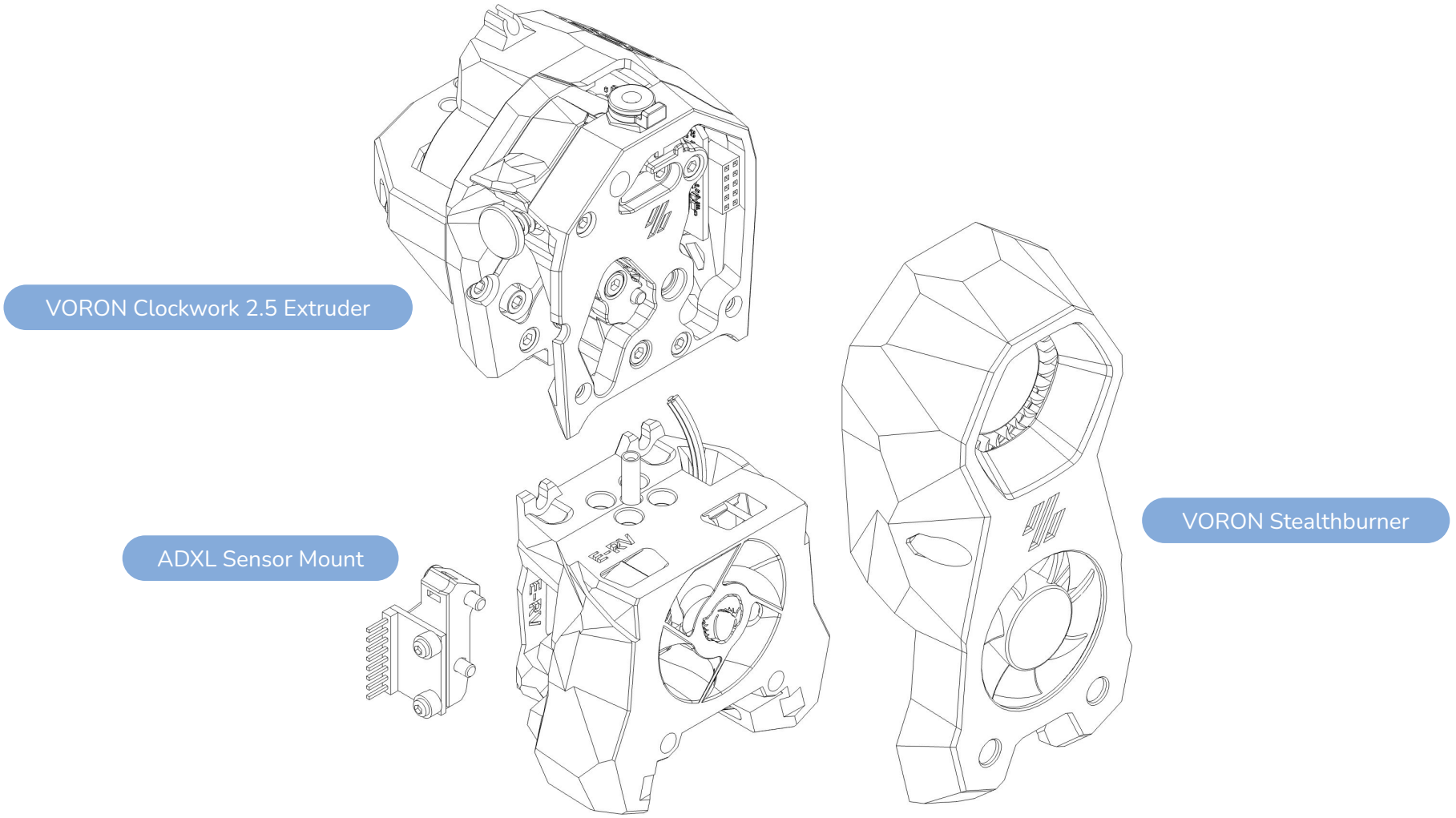
HEX NUT

Hex nuts couple with screws to create a tight, secure joint. Only M3 variants are used throughout this guide.

ISO 4032



8


VORON Clockwork 2.5 Extruder
ADXL Sensor Mount
VORON Stealthburner
VORON Stealthburner Tool Cartridge

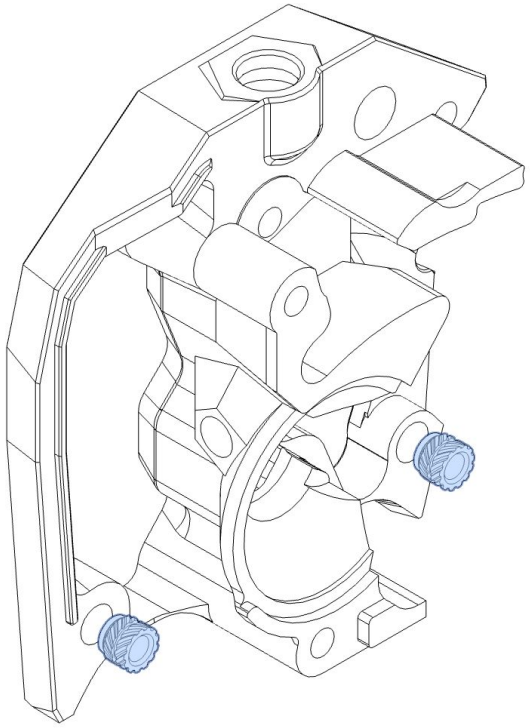


HEAT SET INSERTS

This design relies heavily on heat set inserts. Make sure you have the proper inserts (check the hardware reference for a close-up picture and the Sourcing Guide for dimensions).

If you've never worked with heat set inserts before we recommend you watch the linked guide.

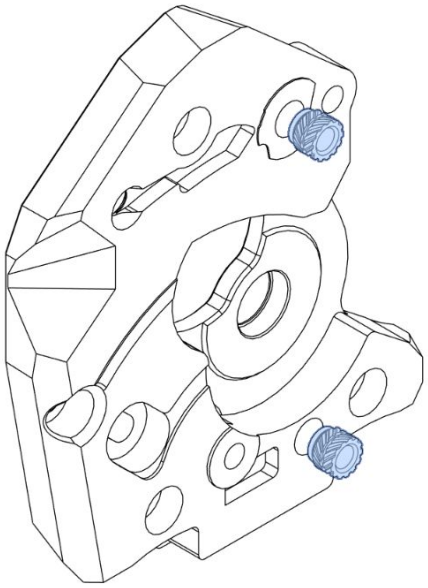
https://voron.link/m5ybt4d



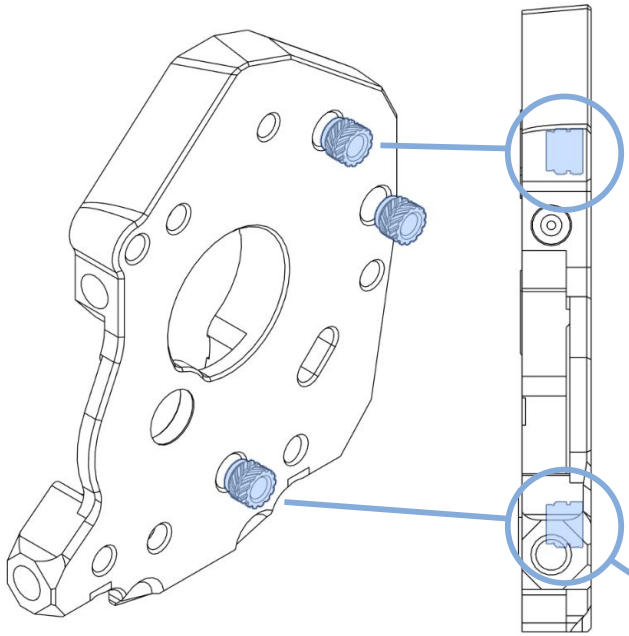
main_body.stl

WHICH BODY?

There are two main body options for Clockwork 2.5 - one that uses an ECAS04 coupler and one that uses a Bowden Collet. For simplicity, the collet version will be shown when the part is otherwise identical.



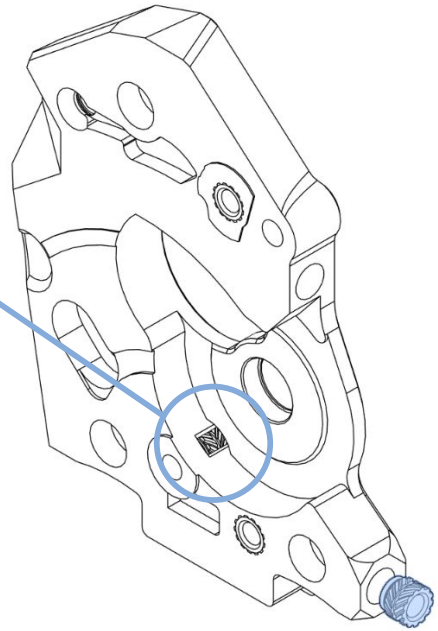
motor_plate.stl



BELOW SURFACE
The top heatset insert needs to sit below the surface of the printed part.

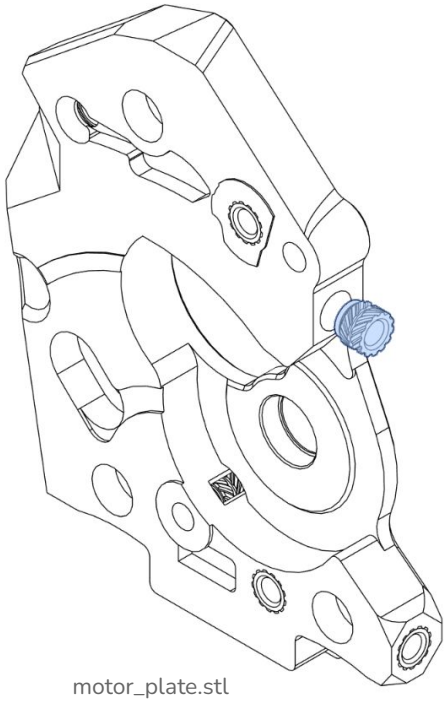
BELOW SURFACE
The bottom heatset insert needs to sit below the surface of the printed part.

Mind the cutout in the part and make sure to install it straight.

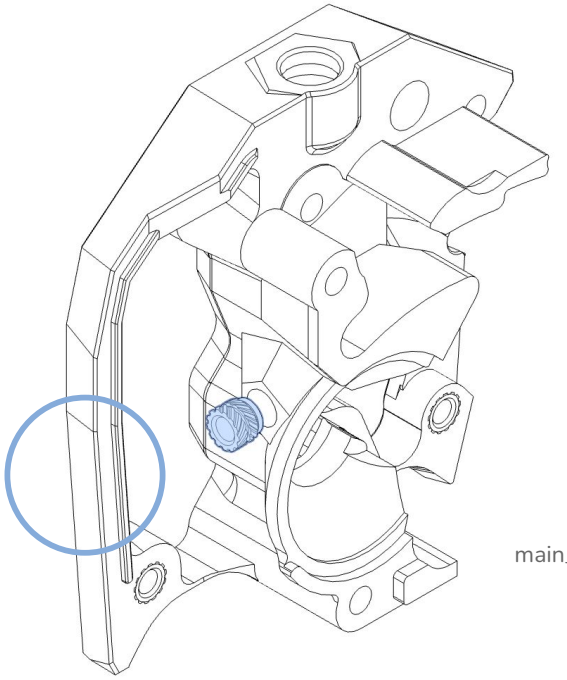


OPTIONAL: TOOLHEAD PCB

If you opt to use a toolhead PCB, add additional heat set inserts into locations highlighted below.



motor_plate.stl



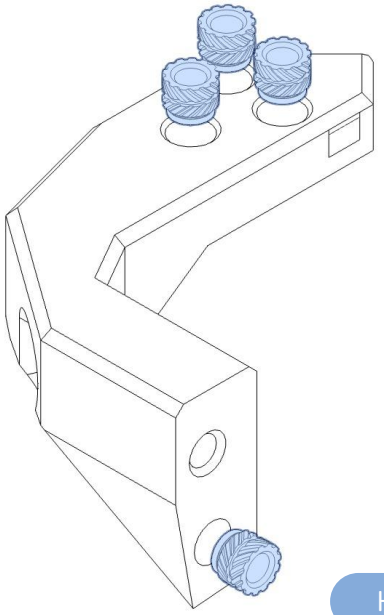
main_body.stl

HEAT SET INSERT

Be careful when inserting this heat set insert. It's easy to accidentally touch the left side of the part with the soldering iron.

GENERIC CABLE CHAINS

The 3-hole pattern is usually found on generic cable chains.

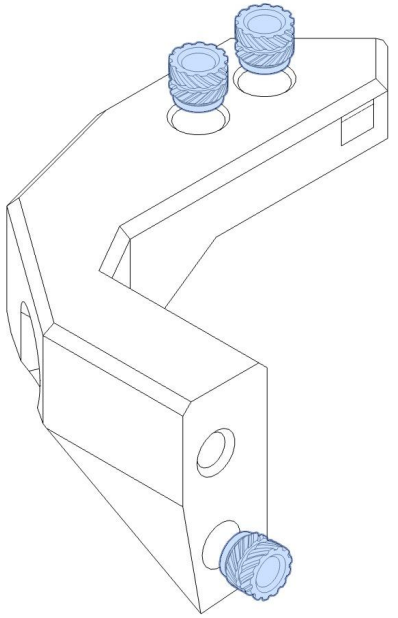


Heat Set Insert

chain_anchor_3hole.stl

IGUS-style CABLE CHAINS

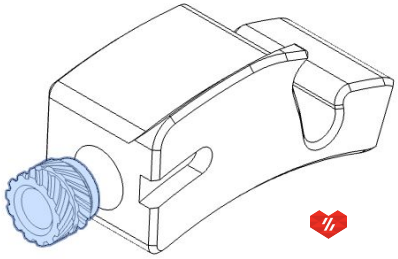
IGUS-style chains have two mounting holes.



chain_anchor_2hole.stl

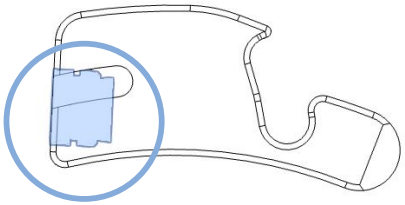
PRINTER SPECIFIC MOUNTS

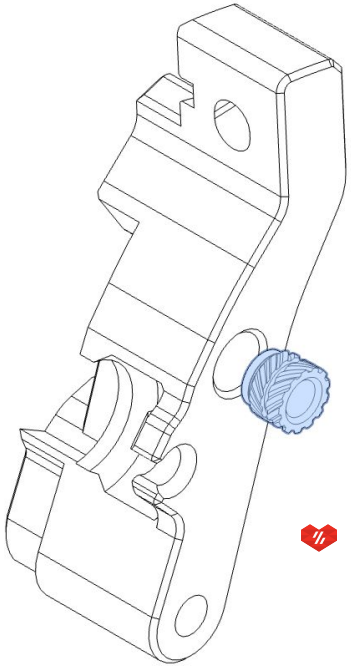
V2, Trident and Legacy use the same printed parts. Extra parts are included for Switchwire.



[a]_latch_shuttle.stl

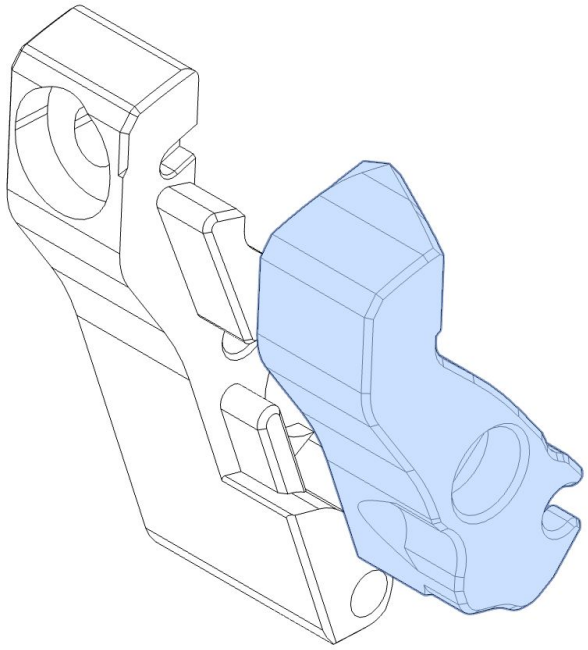
FLUSH WITH SURFACE
The heat set insert needs to sit flush or slightly below the surface of the printed part.



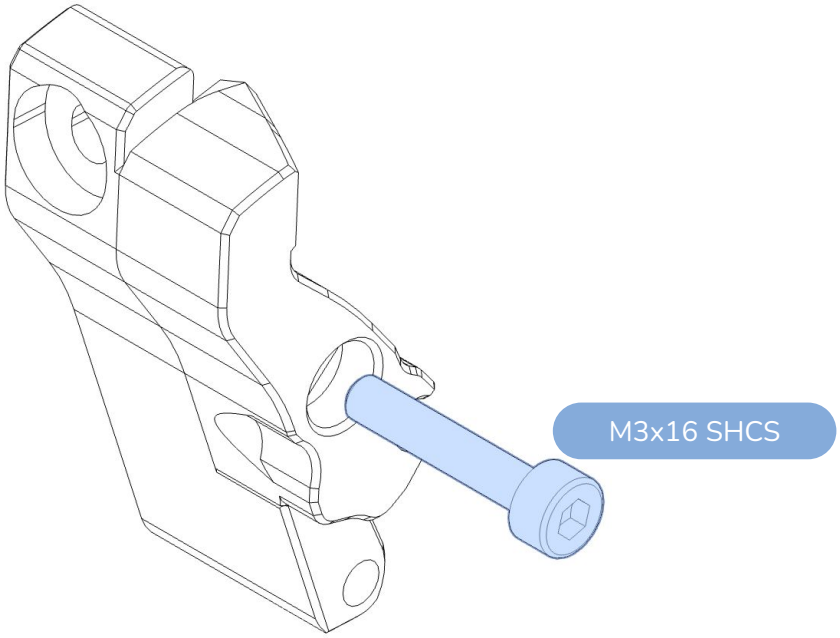


[a]_guidler_a.stl

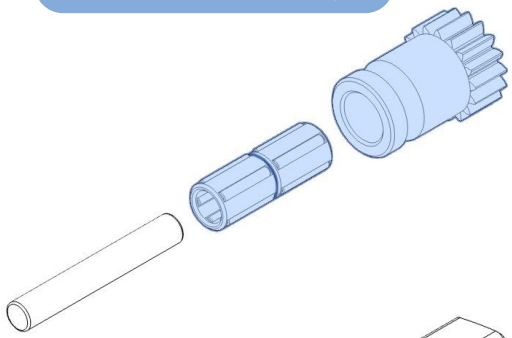
ACCENT PART?
Look for Voron heart next to the part. It indicates that this is a part that is usually printed in the accent color.

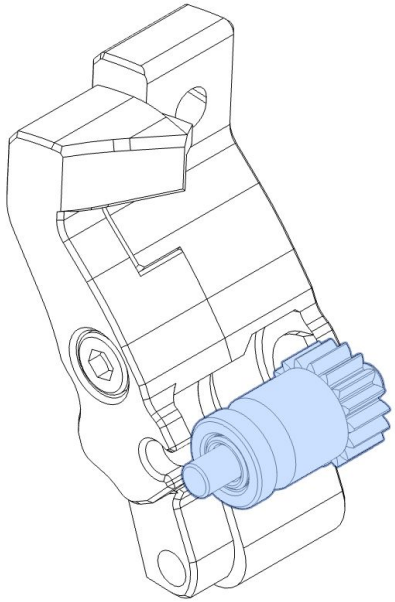


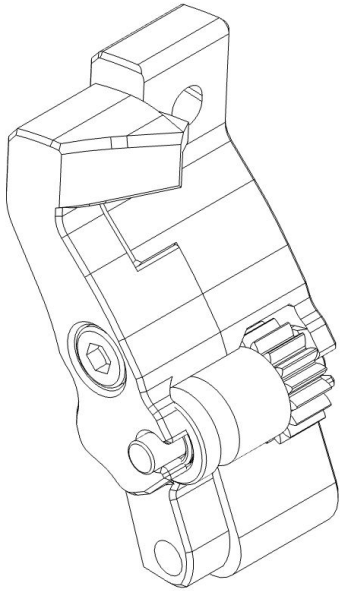
[a]_guidler_b.stl



BMG Idler Assembly

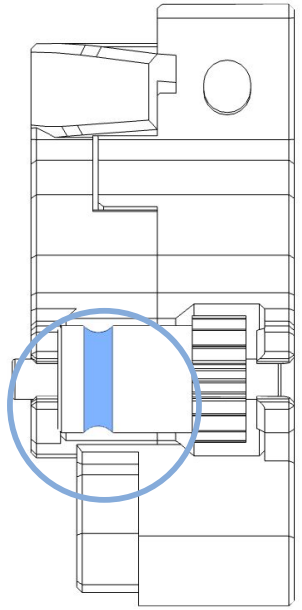






LUBRICATE BEARINGS

A lubrication film is required on the needle bearings to ensure smooth operation and longevity. Refer to the Voron sourcing guide for lubricant options - look for a “light grease”.



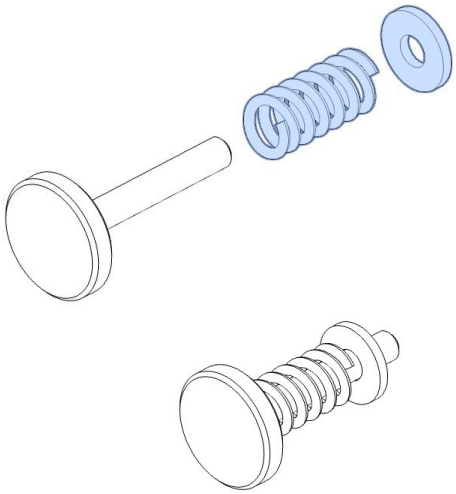
MIND ORIENTATION

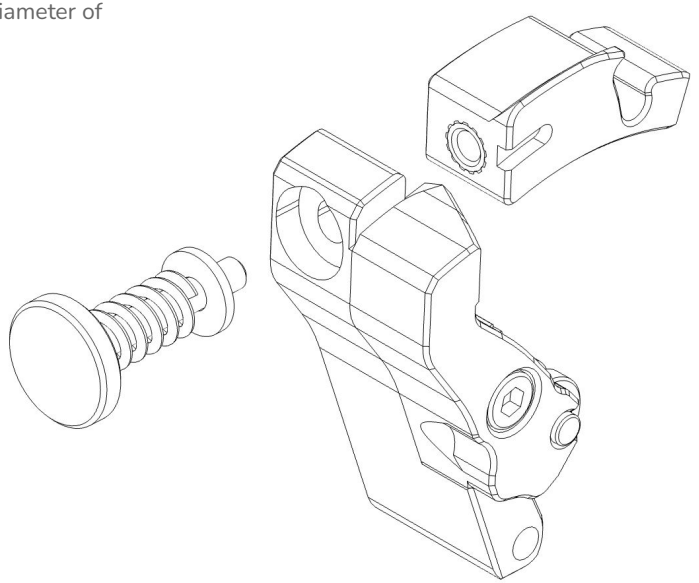
Make sure to orient the idler assembly as shown above.

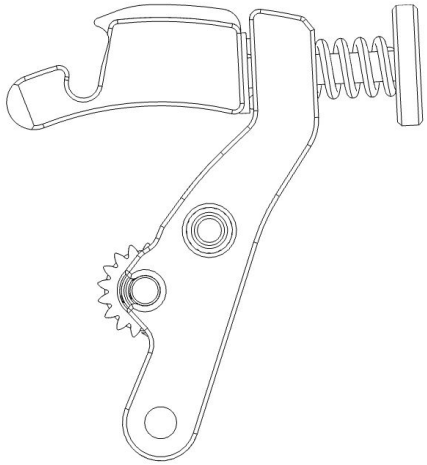
A NOTE ON SPRINGS

Longer/shorter/stiffer springs will change the tension characteristics and have an impact on how well the tension mechanism works. Consider buying the original Bondtech part as those are known to work well. If sourced from a different vendor check if it's roughly 12mm long with an outer diameter of 6mm and a wire thickness of 1mm.

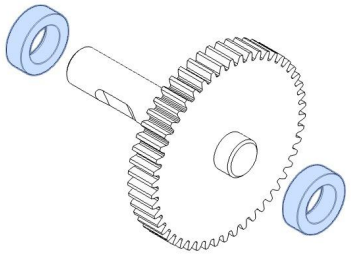
BMG Thumbscrew Assembly







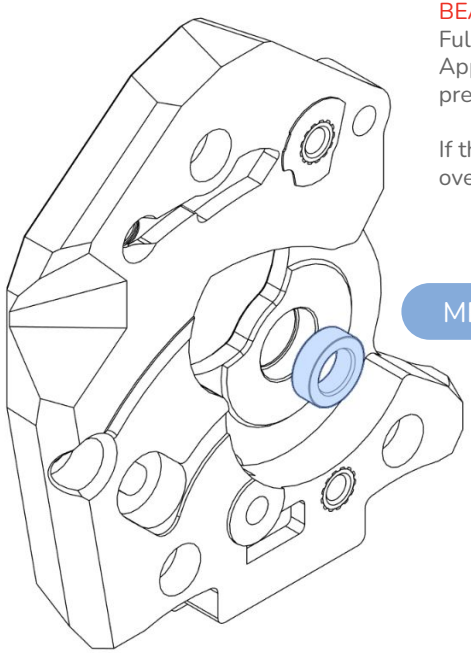
MR85 Bearing



CHECK BEARING FIT
The bearings must slip on and off the shaft easily to allow the gear to self-centre.

Pressing the bearings on the shaft will damage them.

Lightly sand the shaft if required.

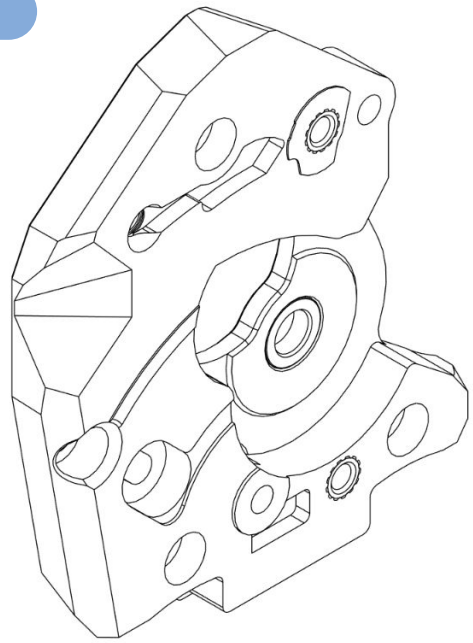


motor_plate.stl

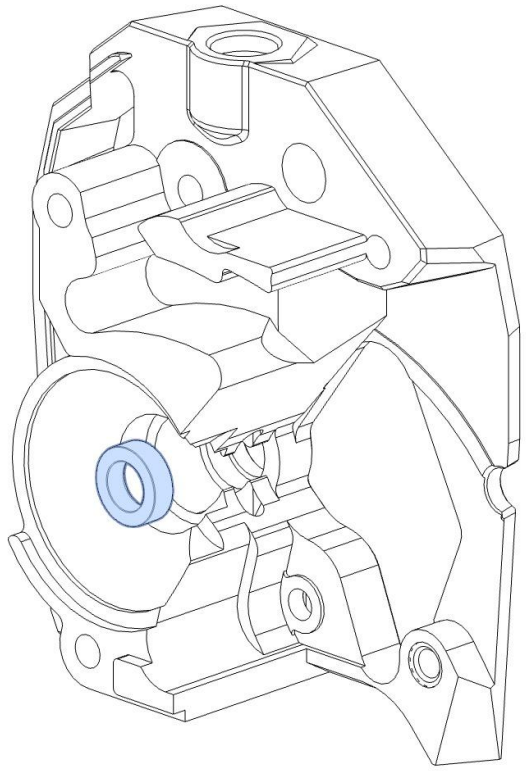
BEARING FIT
Fully seat the bearing into the plastic pocket. Apply even pressure to insert them. Avoid pressing on the inner ring of the bearing.

If the fit is too tight the printed parts are likely over-extruded.

MR85 Bearing



MR85 Bearing

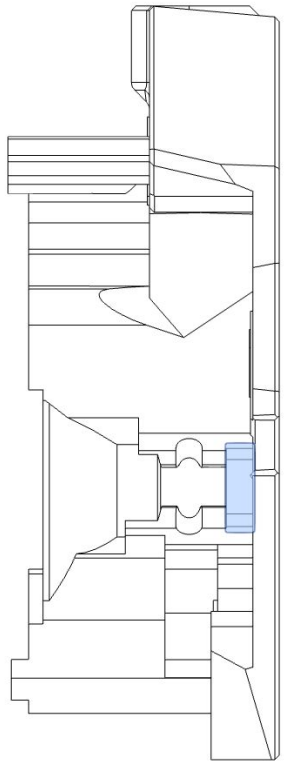


main_body.stl

BEARING FIT

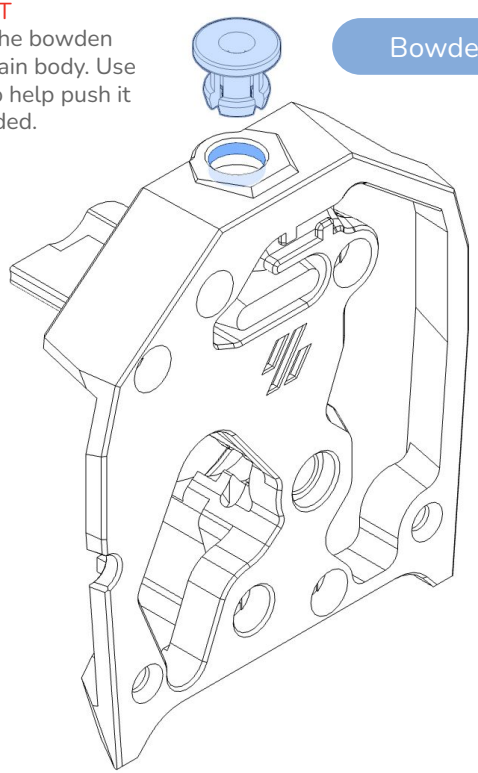
Fully seat the bearing into the plastic pocket. Apply even pressure to insert them.

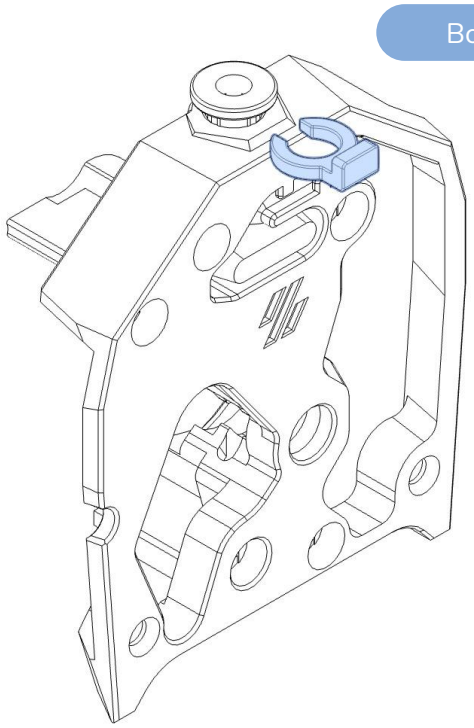
Avoid pressing on the inner ring of the bearing.



CHOICES, CHOICES
There are two options for the PTFE entry retainer - a Bowden Collet (This one) and ECAS04 (next page).

INSTALL COLLET
Press fit install the bowden collet into the main body. Use a hard surface to help push it into place if needed.

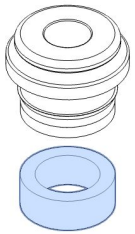




INSTALL CLIP
Install the Bowden clip to secure the collet and prevent the PTFE tube from moving during extrusion and retraction.

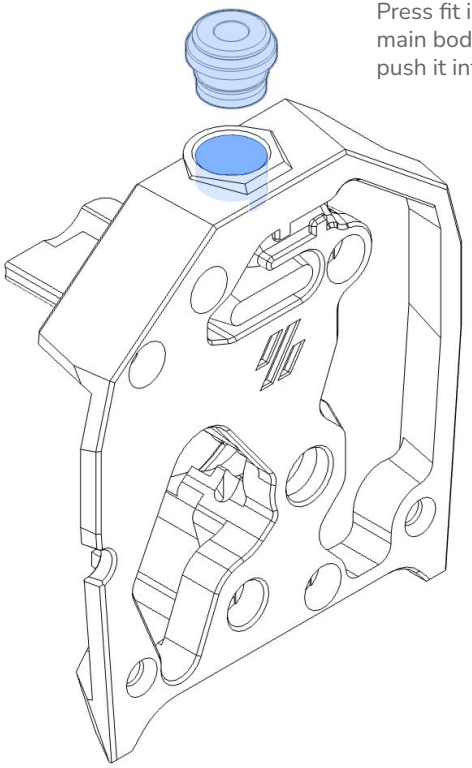
To remove the PTFE tube, first remove the Bowden clip.

ECAS04 Coupling



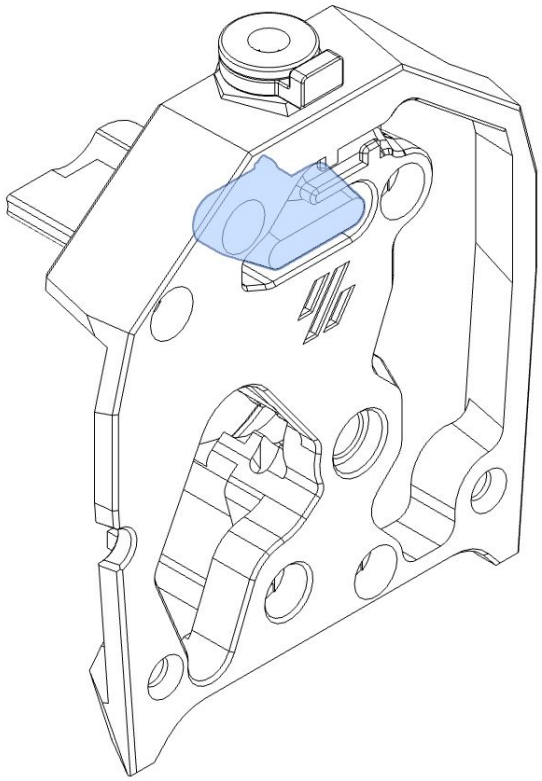
REMOVE RUBBER GASKET

The rubber gasket included with most ECAS04 fittings is not required. Remove and discard appropriately.



INSTALL ECAS04

Press fit install the ECAS04 fitting into the main body. Use a hard surface to help push it into place if needed.

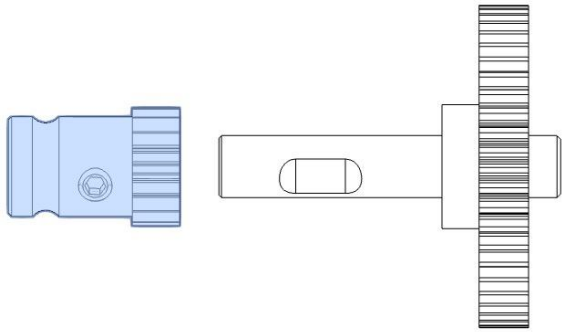


REMOVE SUPPORT

Remove the support with a pair of needle nose pliers, or by pushing it out through the screw hole from behind with a hex key.

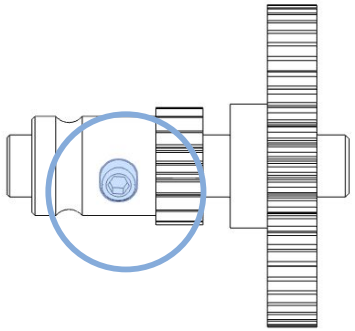
A NOTE ON GEARS

Poorly made gears often cause print quality issues. For best performance consider sourcing the original Bondtech parts.



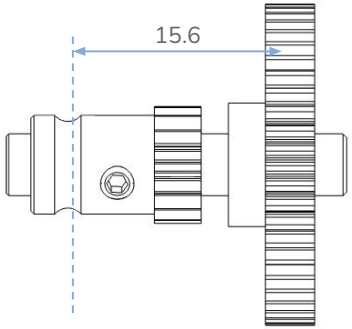
DRIVE GEAR

Make sure the set screw in the filament drive gear is seated against the notch in the shaft. Carefully tighten the set screw, the head is easy to strip.



INITIAL POSITION

The final position is set later.

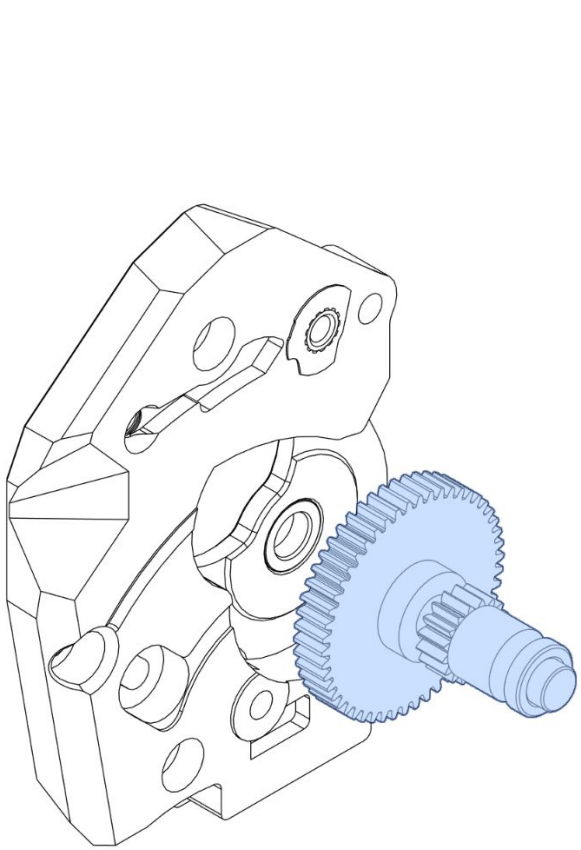


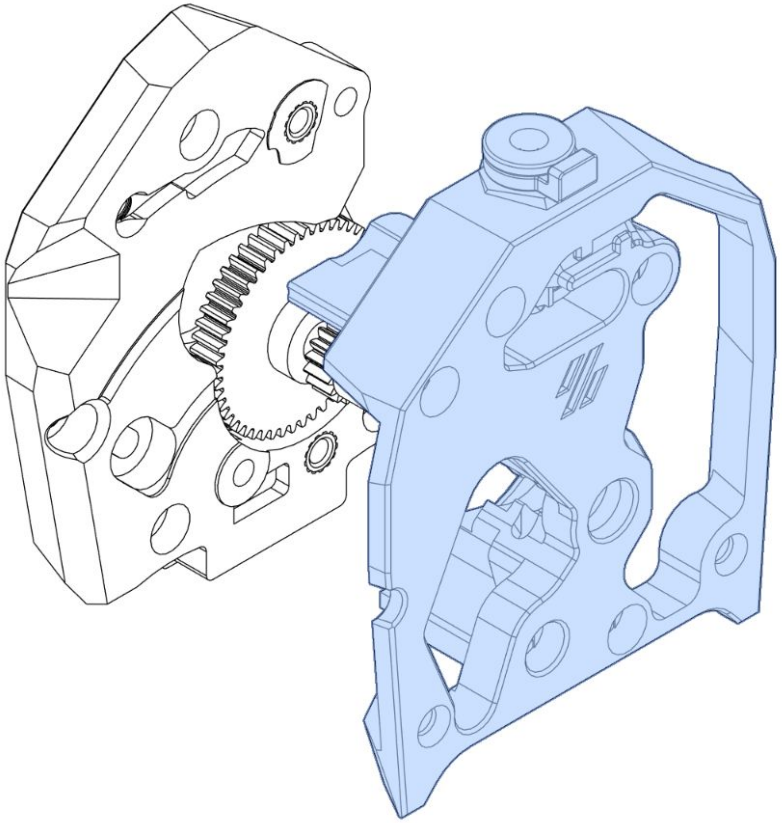
THREAD LOCKER

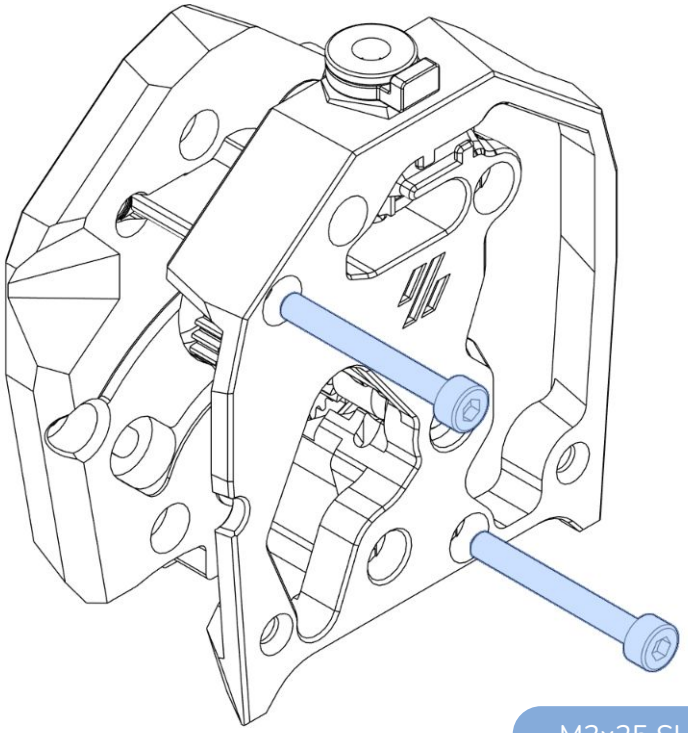
The final position of the drive gear is set in a later step. Common thread lockers have a long enough working time to complete the steps without issues.

Familiarize yourself with the steps on the next 3 pages before you apply thread locker.

Complete the steps on the next 3 pages after applying the thread locker.

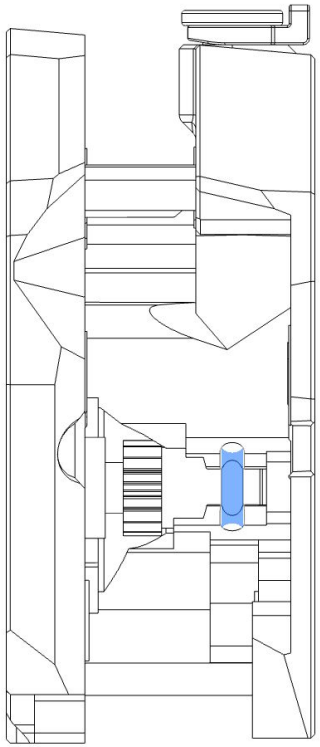




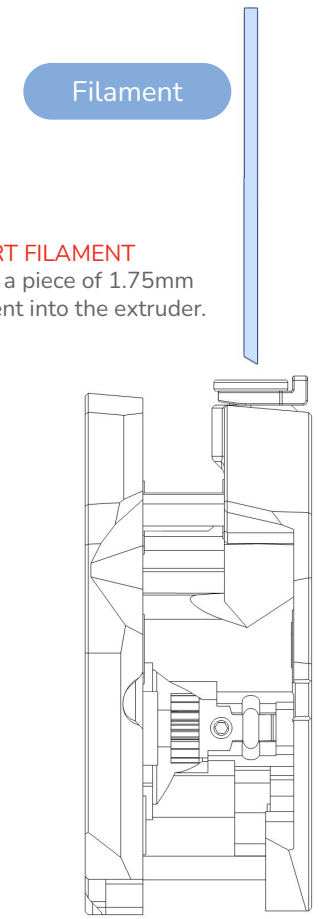


M3x25 SHCS

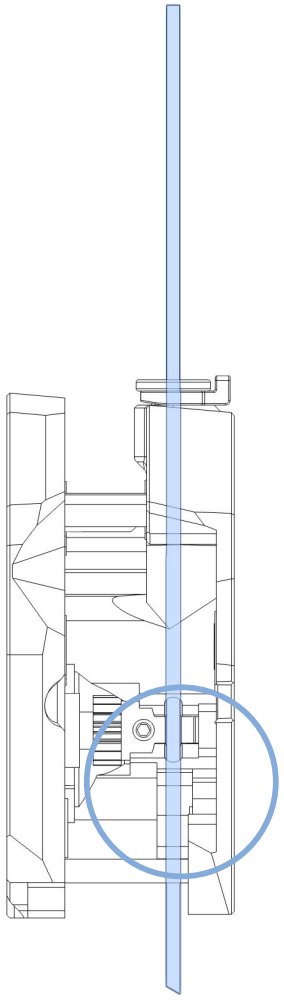
DON'T OVERTIGHTEN
Tighten until the plastic bends and cracks. Back up 2 turns, discard parts, reprint and try again.



INITIAL ALIGNMENT CHECK
Check if the filament path aligns with the toothed section of the drive gear.



INSERT FILAMENT
Insert a piece of 1.75mm filament into the extruder.



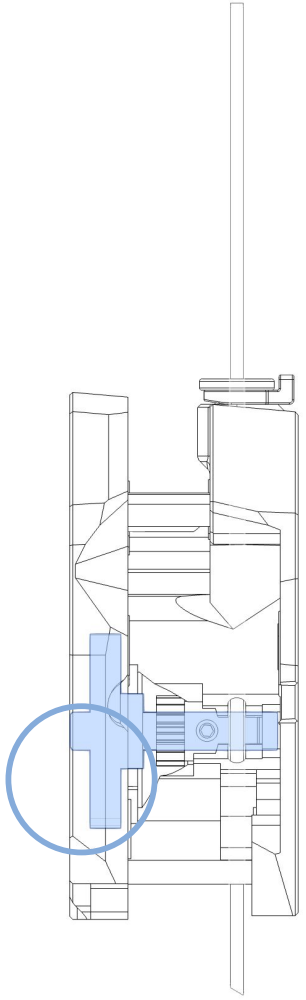
CHECK ALIGNMENT
With the filament inserted, verify if the filament path and drive gear are aligned.

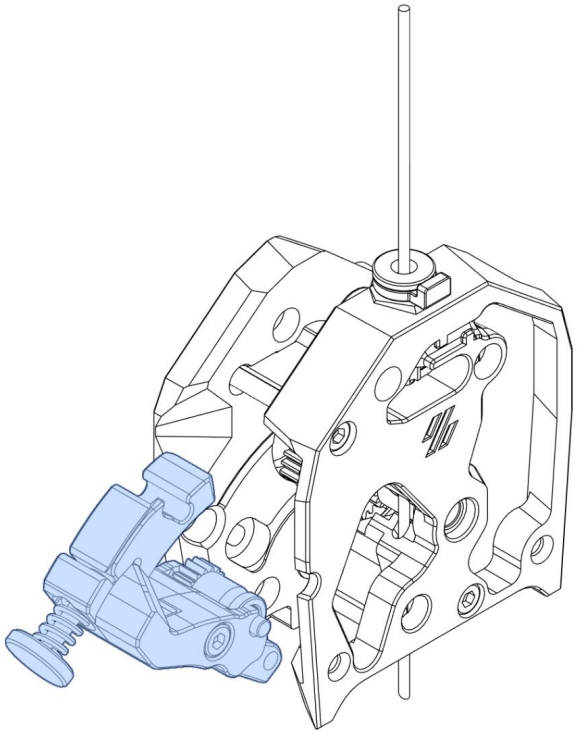
Loosen the set screw and adjust the position of the drive gear if required.

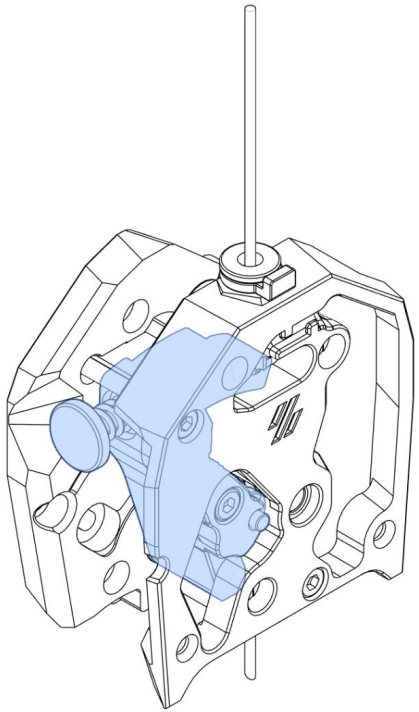
CHECK FOR CLEARANCE, CLARENCE
The drive shaft must not touch the motor housing. Make sure it does not sit above the surface of the printed part.

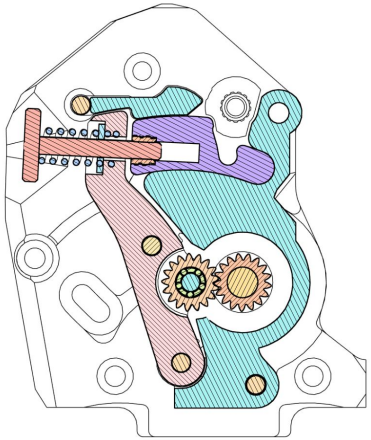
Check if the shaft has sufficient clearance when fully seated.

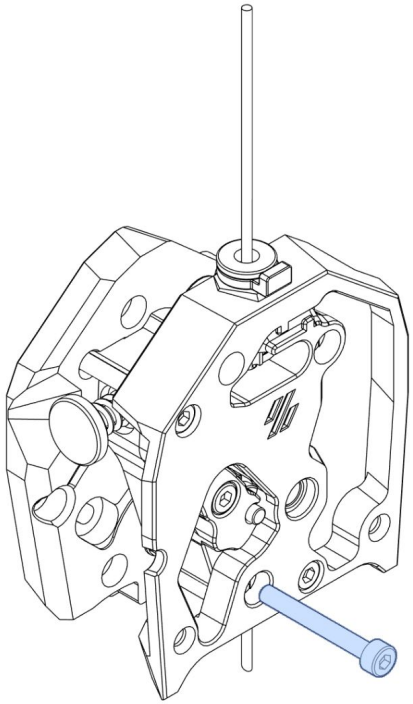
Depending on the shaft's tolerances you may need to adjust the position of the drive gear or sand the face of the shaft.







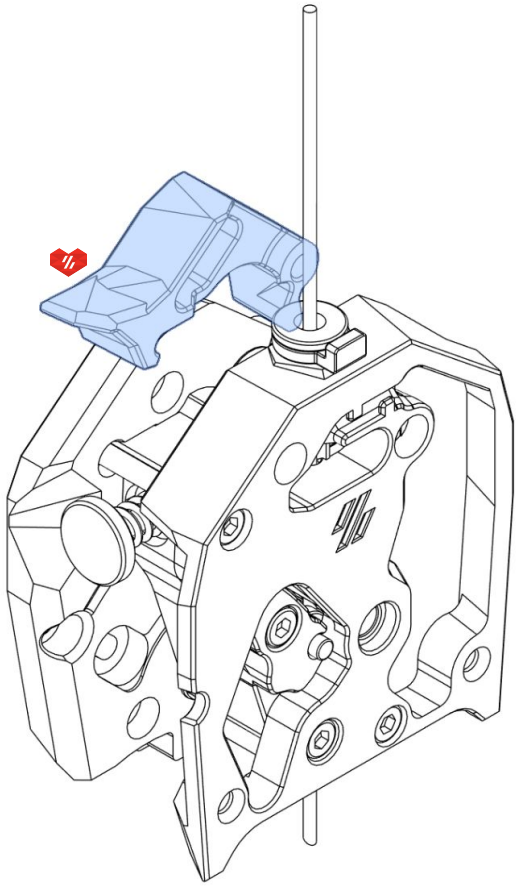




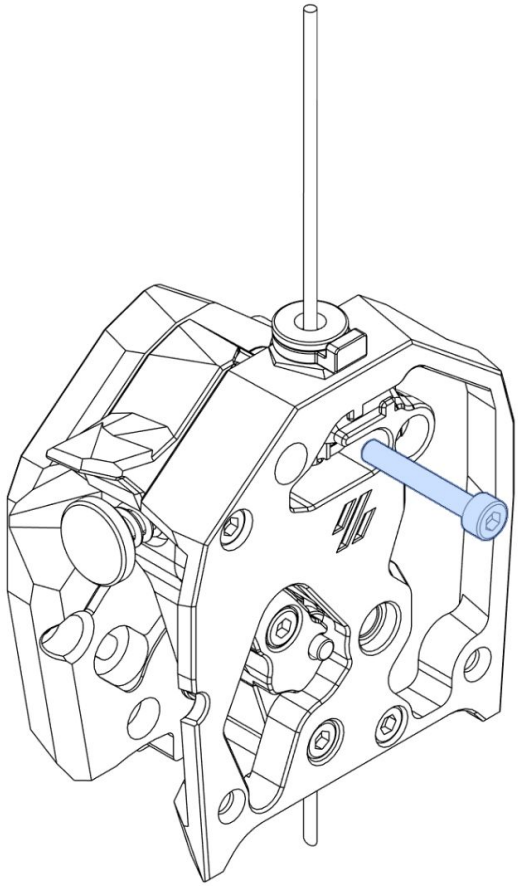
M3x25 SHCS

DON'T TIGHTEN

The tensioning arm must move freely.
Tightening the bolt may lock it in place.



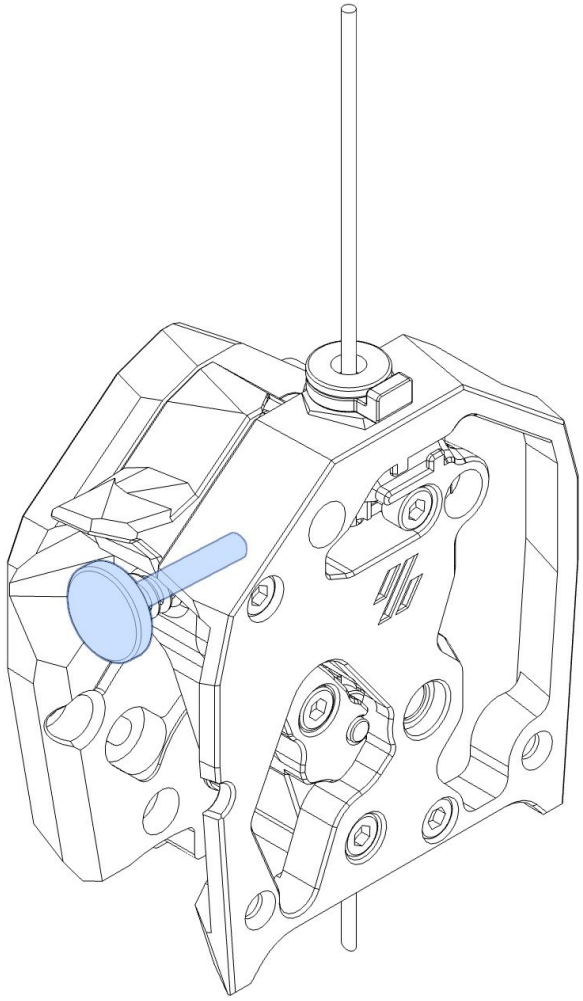
[a]_latch.stl

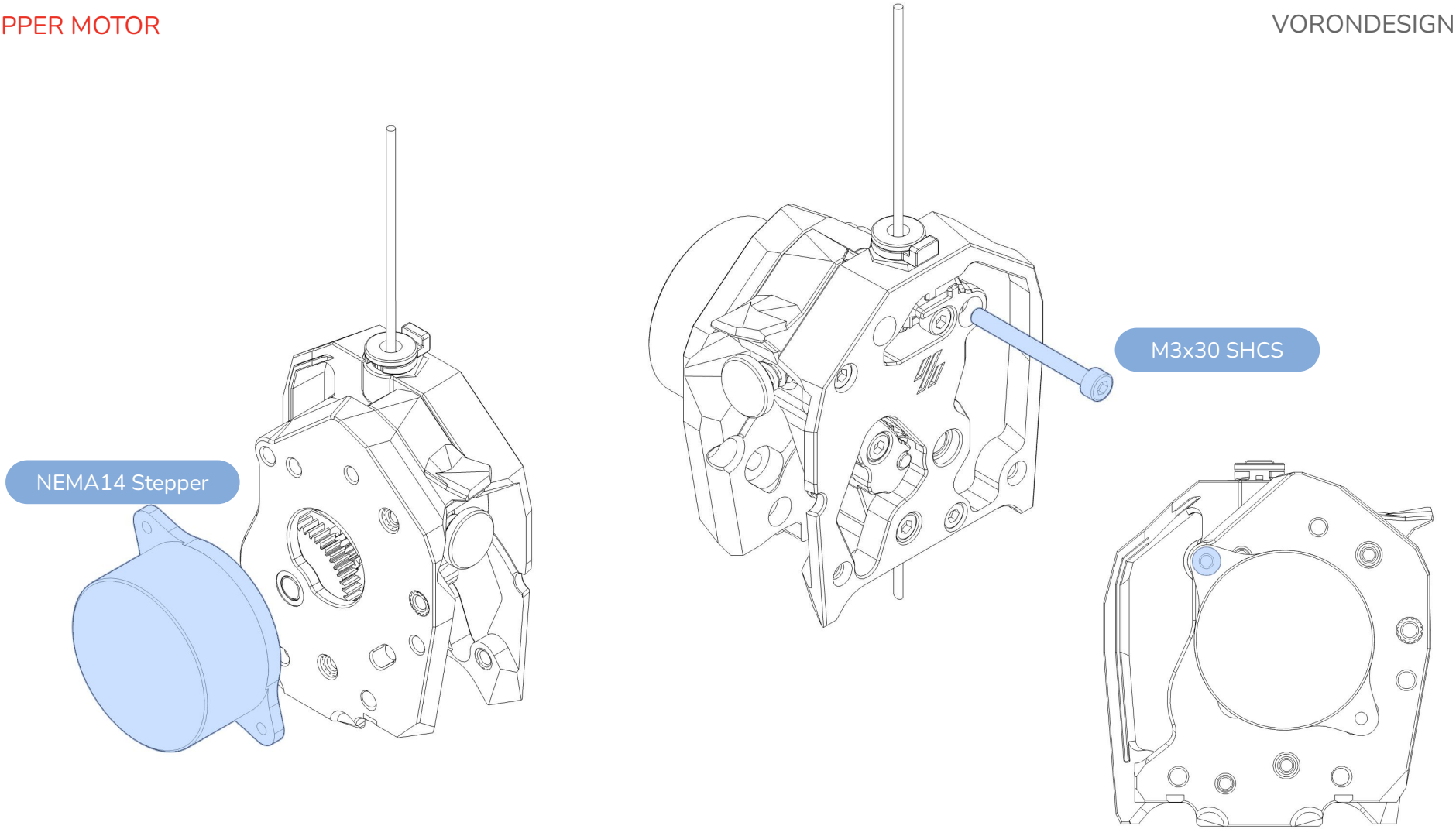


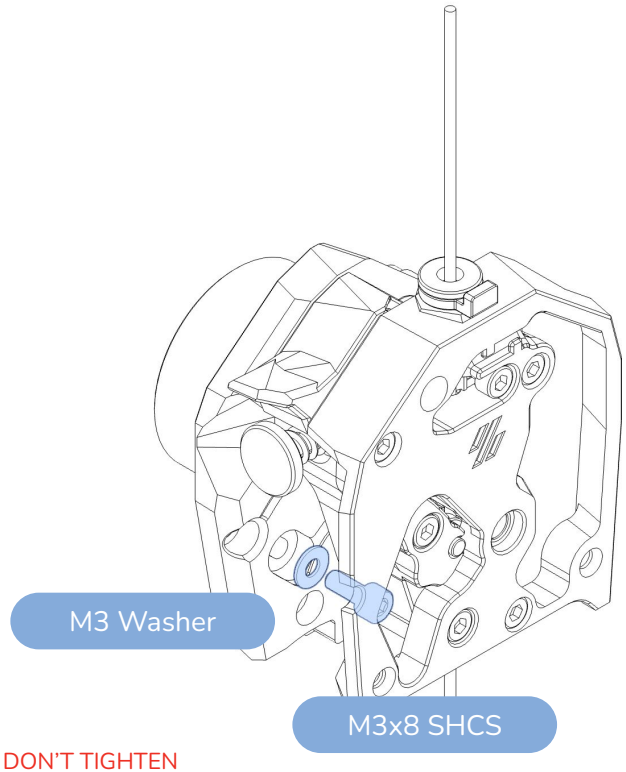
M3x20 SHCS

TENSION KNOB

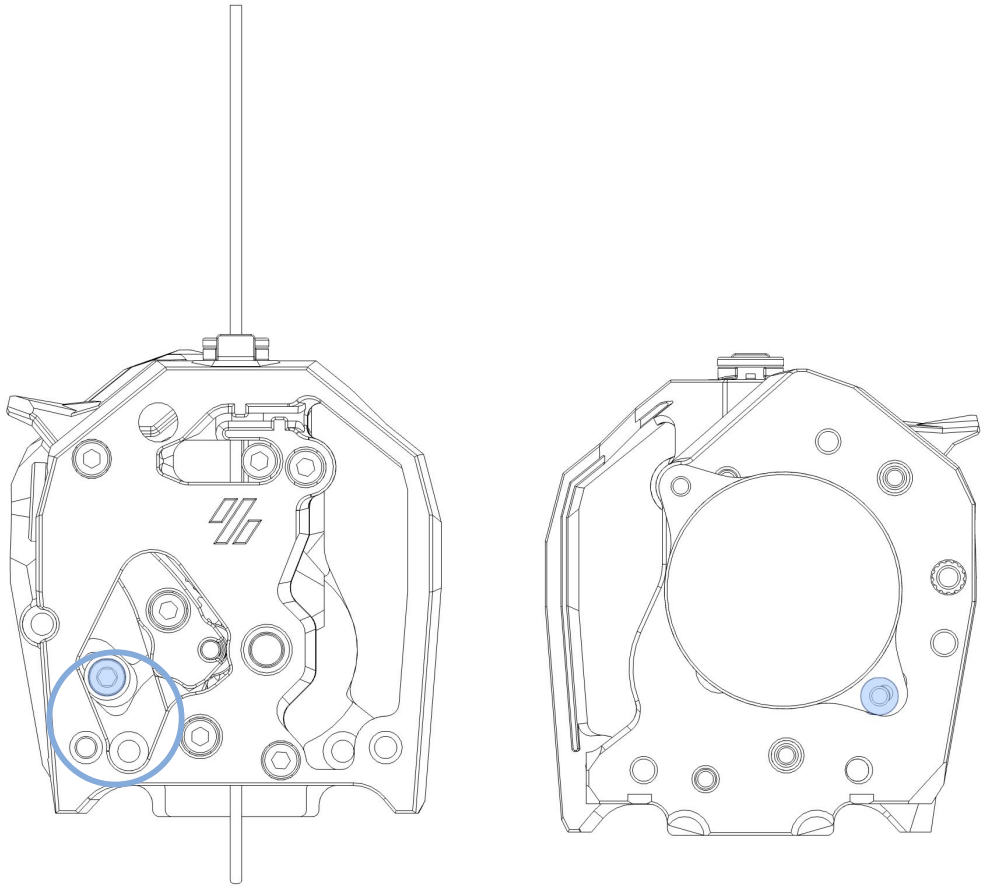
Turning the thumb screw clockwise will increase the tension and grip on the filament. Too much tension will result in print issues.



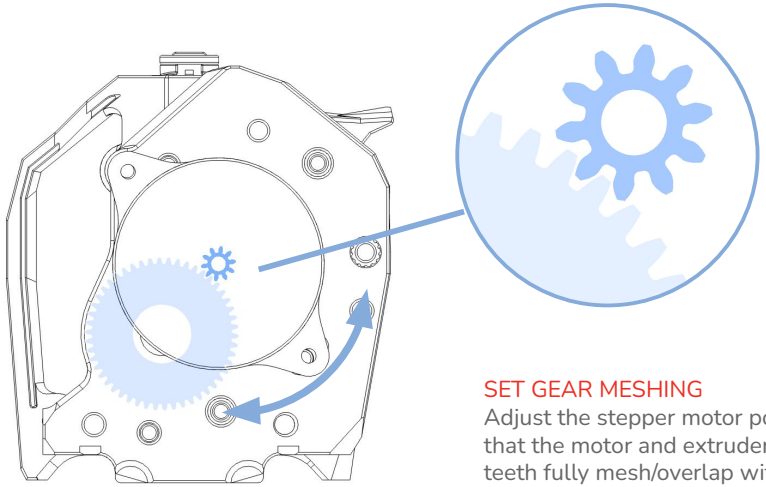

NEMA14 Stepper
M3x30 SHCS



DON'T TIGHTEN
The motor position will be adjusted in the next steps. Don't fully tighten just yet.

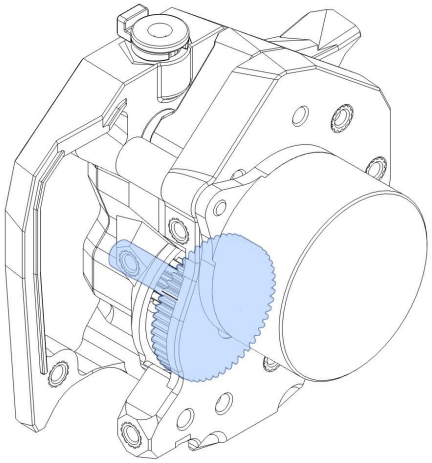


ACCESS HOLE
The second motor bolt can be accessed from the front of the extruder.



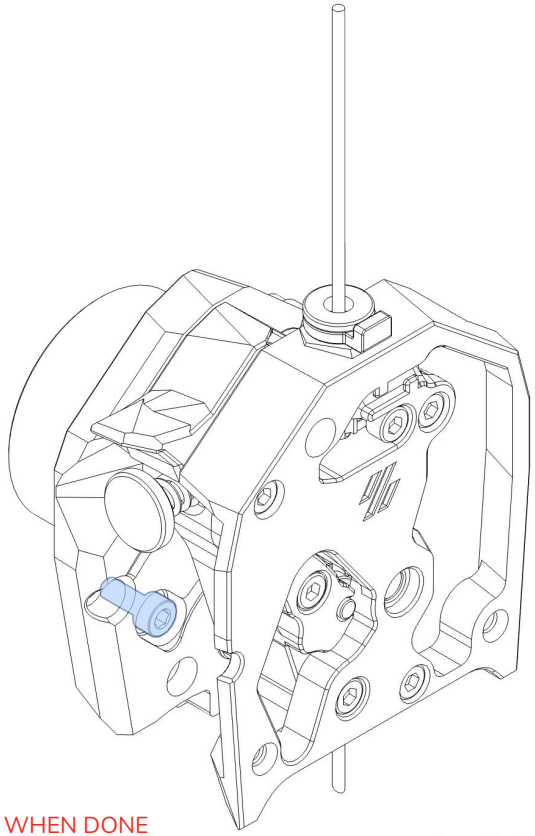
SET GEAR MESHING

Adjust the stepper motor position so that the motor and extruder gear teeth fully mesh/overlap with each other. There should be a very small gap between the faces of the gear teeth; the gears must not press tightly against each other.



CHECK GEAR PLAY

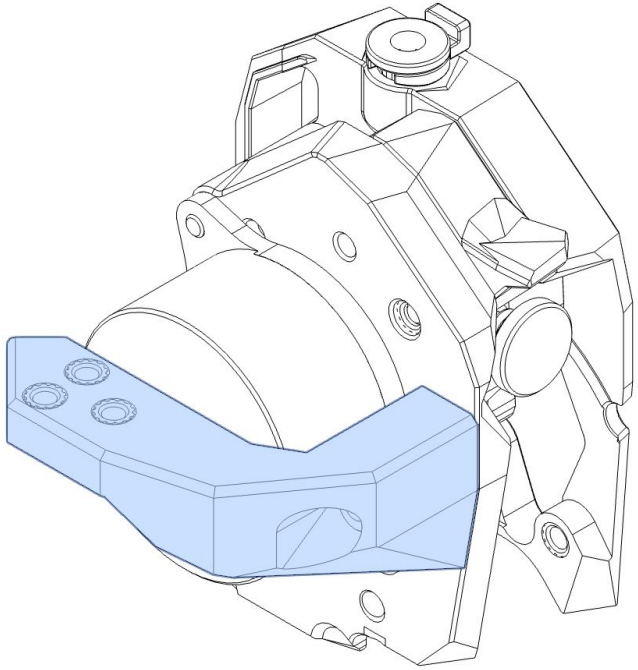
The gear should have a slight play and should not be fully tight against the pinion. Adjust the position of the motor until you have a faint play.



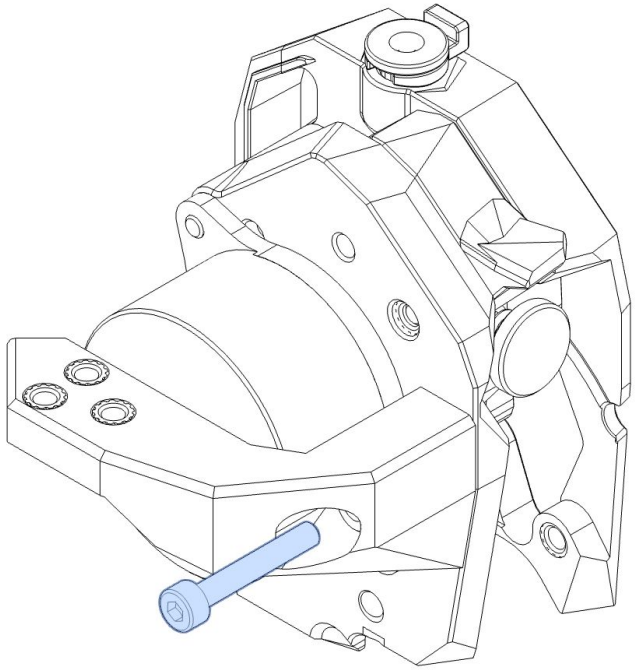
TIGHTEN WHEN DONE

Don't forget to tighten the second motor bolt after adjusting.

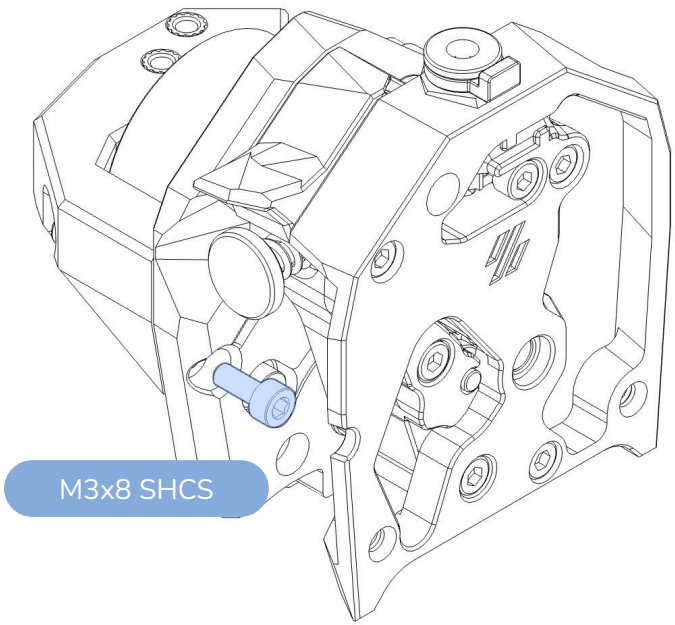
https://voron.link/u6ehxsw

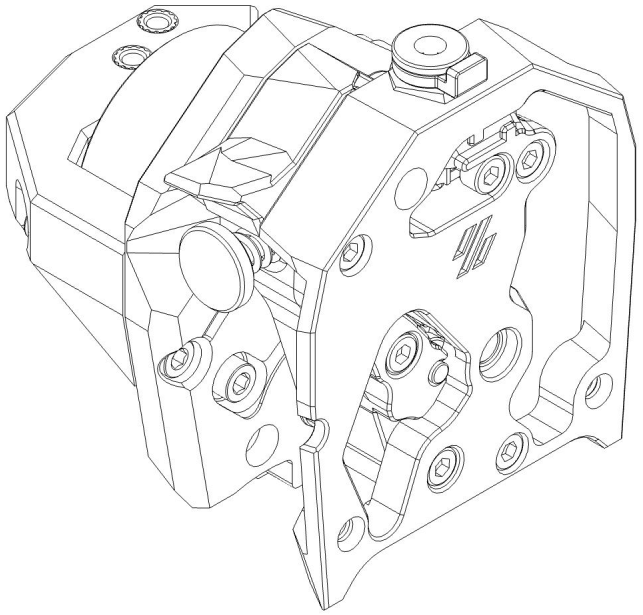


GENERIC CHAIN MOUNT
This example shows the generic cable chain mount.
Other chain mounts will look different.



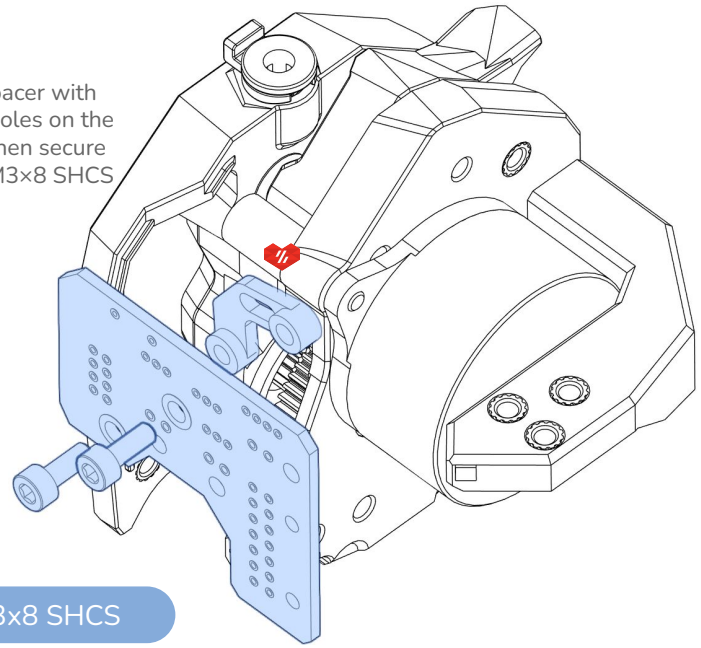
M3x20 SHCS

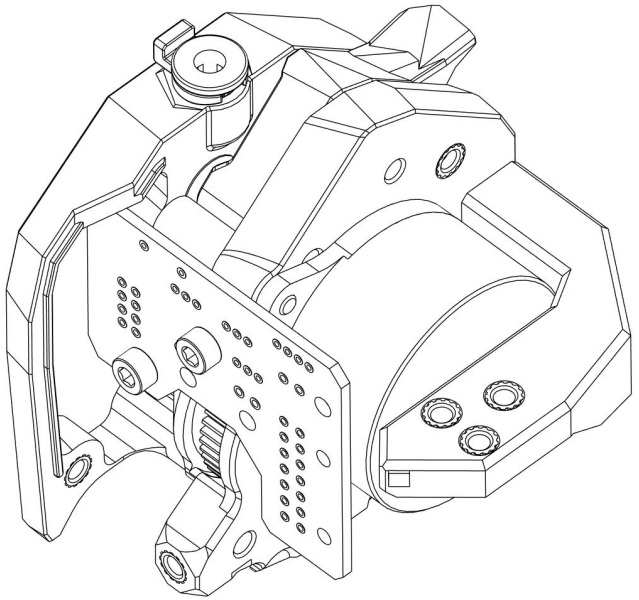

M3x8 SHCS



PCB SPACER

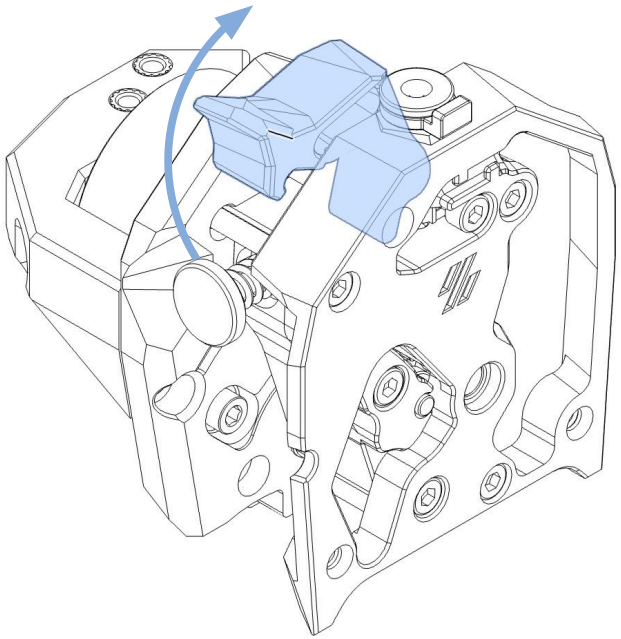
Align the printed spacer with the two mounting holes on the SB toolhead PCB, then secure the PCB with two M3×8 SHCS screws.

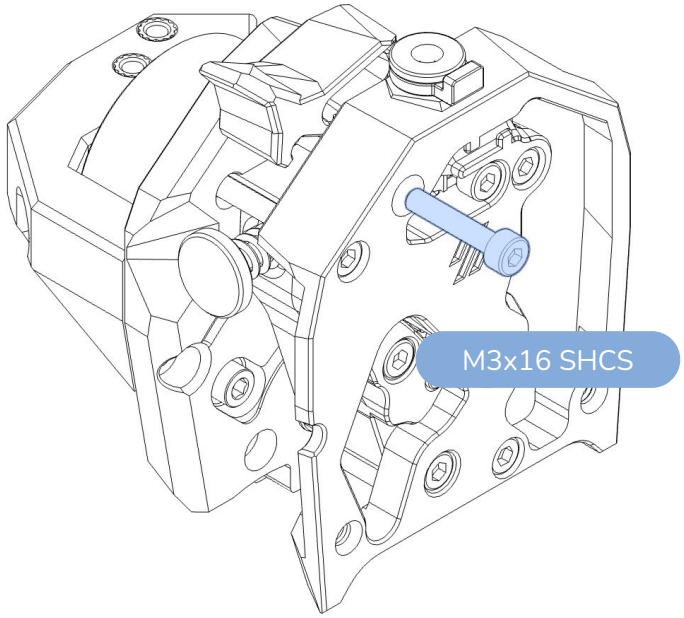


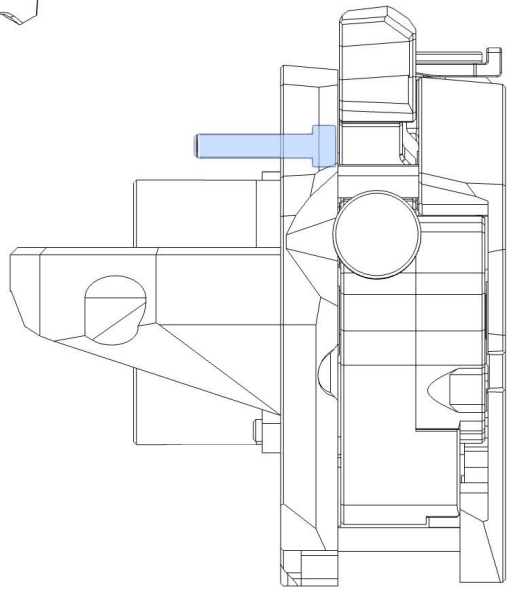


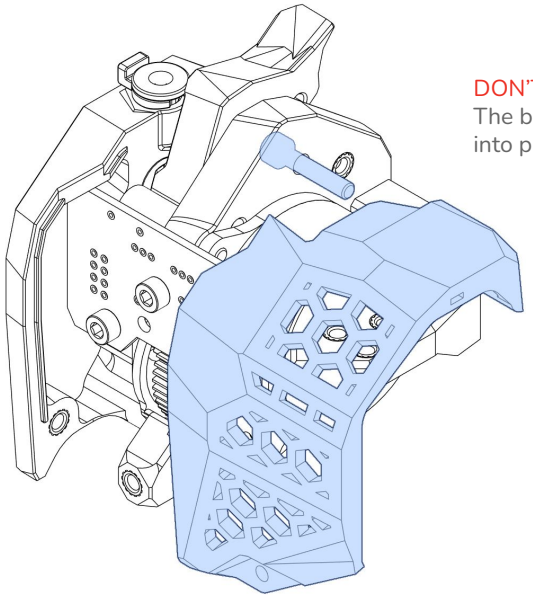
OPEN LATCH

Undo the filament latch to expose the bolt pocket for the cable cover.

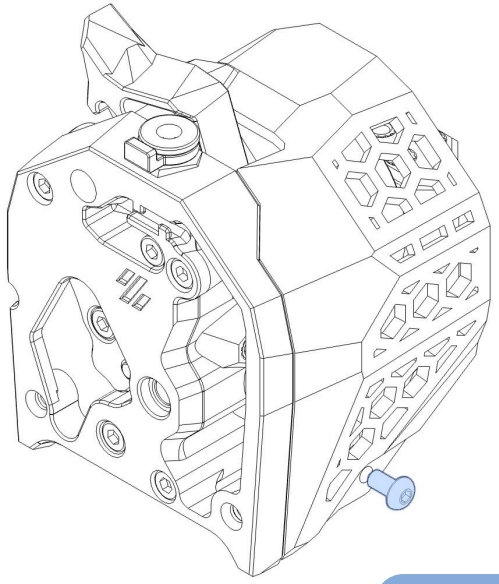




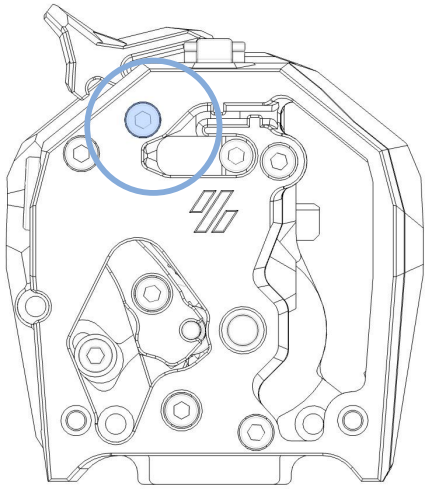




DON'T OVER-TIGHTEN
The bolt is threaded directly into plastic.



M3x6 BHCS



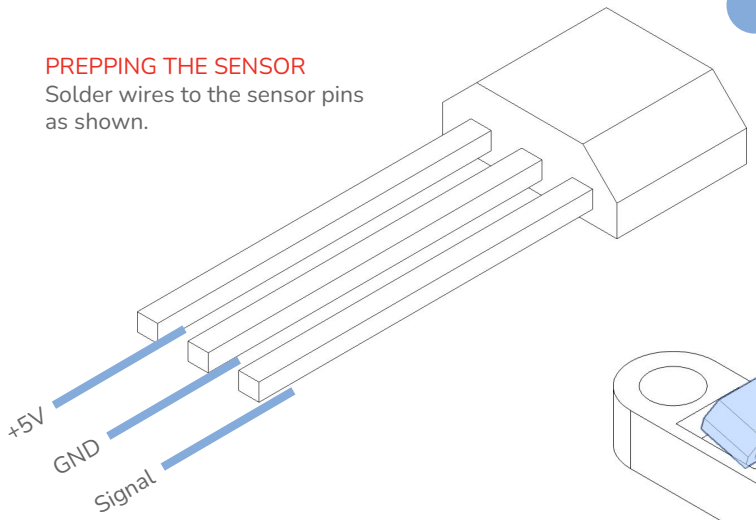
ACCESS HOLE
The bolt can be accessed from the front of the extruder.

HALL EFFECT SENSOR (OPTIONAL)

Hall Effect Sensor

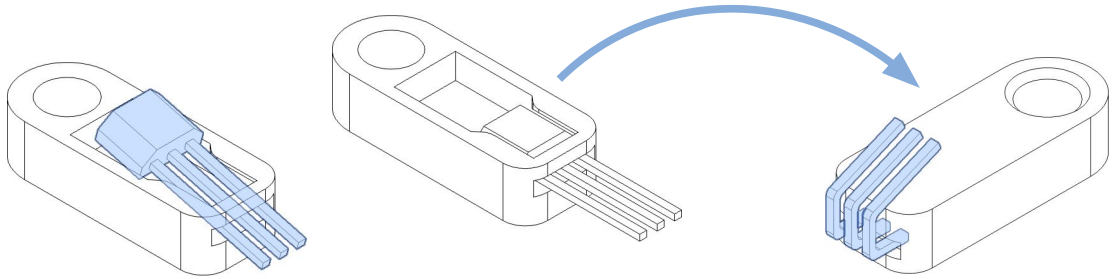
PREPPING THE SENSOR

Solder wires to the sensor pins as shown.



BEND THE PINS

Turn the assembly over and carefully bend the pins around the printed bracket. The pins are delicate and may break if bent repeatedly or too sharply.

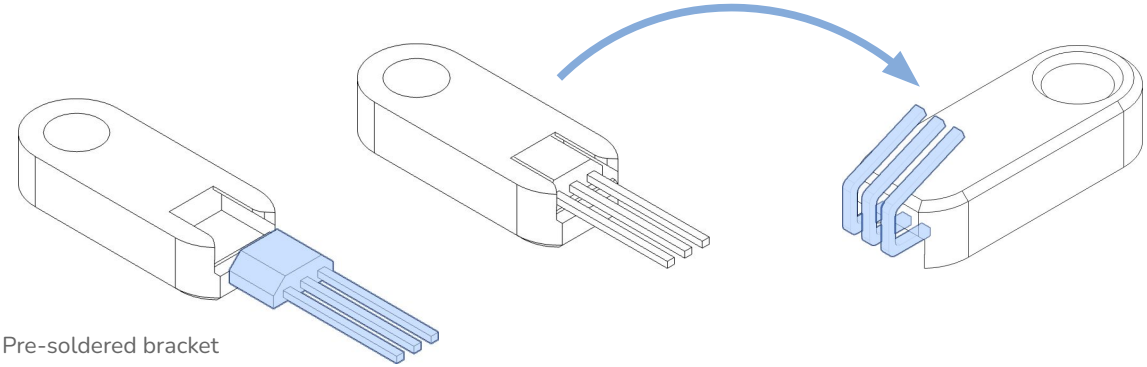


Self-sourced bracket

ASSEMBLE THE SENSOR

Run wires and pins through the hole in the printed bracket, and slide sensor into place.

Two printed bracket versions are available. Use the version that corresponds with your hardware.



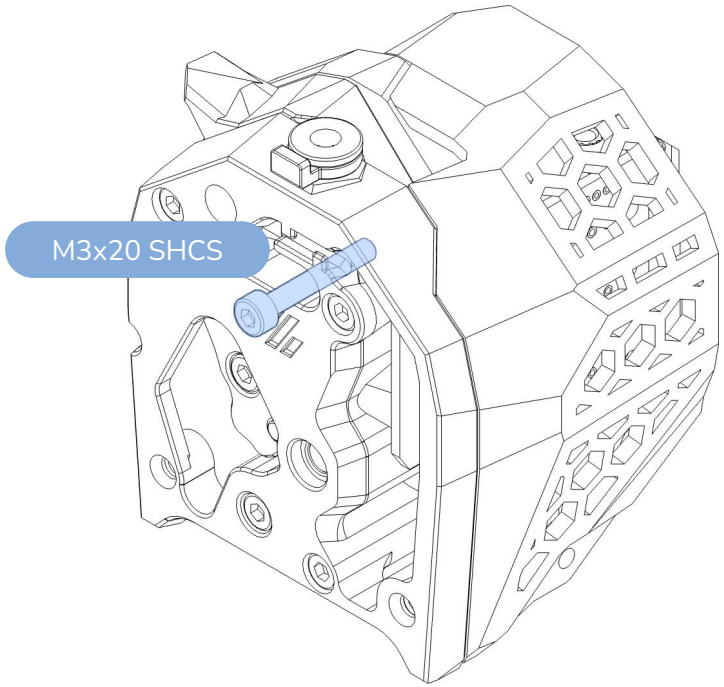
Pre-soldered bracket

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HALL EFFECT SENSOR (OPTIONAL)

REMOVE SCREW

Remove this M3x20 SHCS to allow installation of the hall effect sensor.



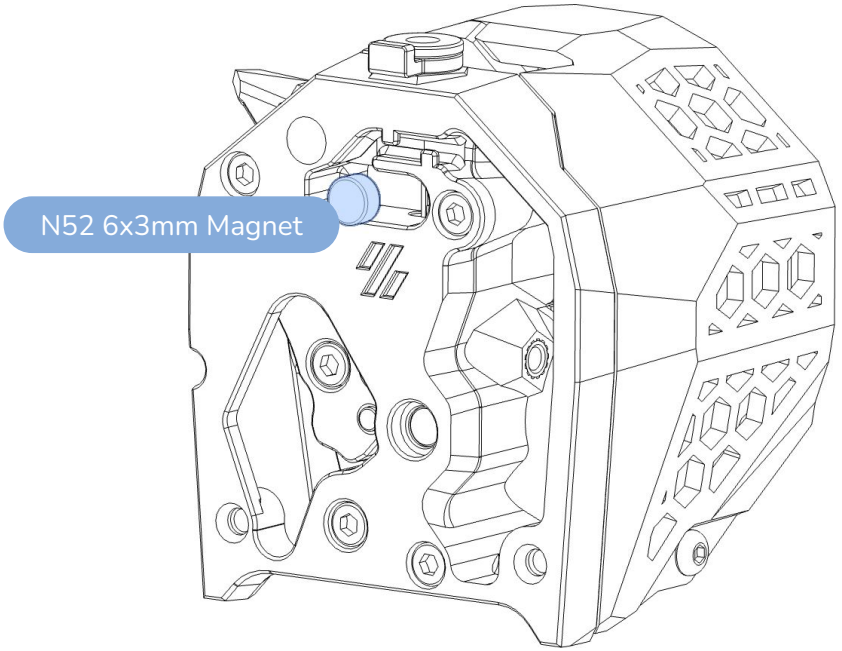
THE N52 MAGNET

Insert the N52 6x3mm magnet into the channel, south pole facing outward. Make sure it can slide horizontally without obstruction.

To determine the south pole, stack two magnets together to strengthen the magnetic field. Then stand the stack on its circumference on a smooth, level surface so it can rotate freely. It should naturally align with Earth's magnetic field, allowing you to identify the north- and south-facing poles.

The linked guide may be helpful for this step.

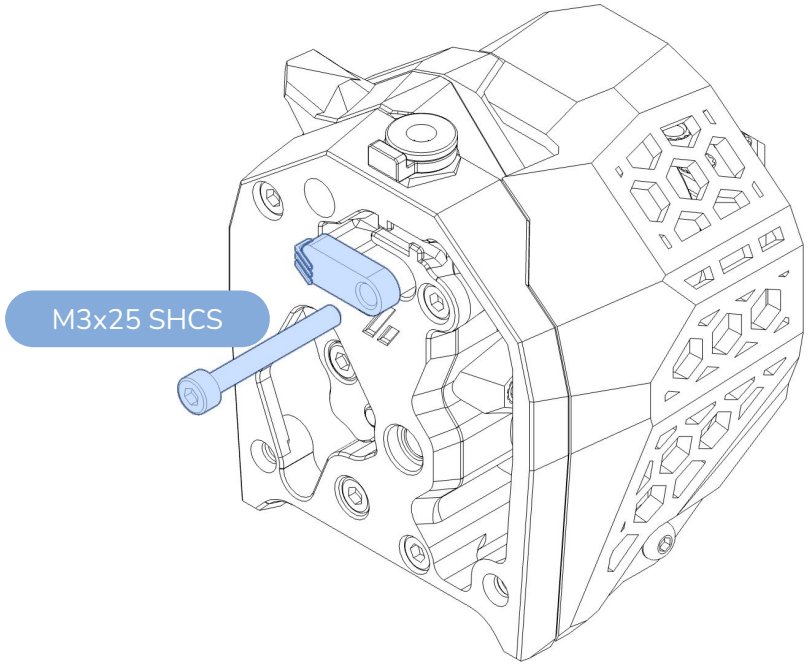
https://voron.link/htekfs8



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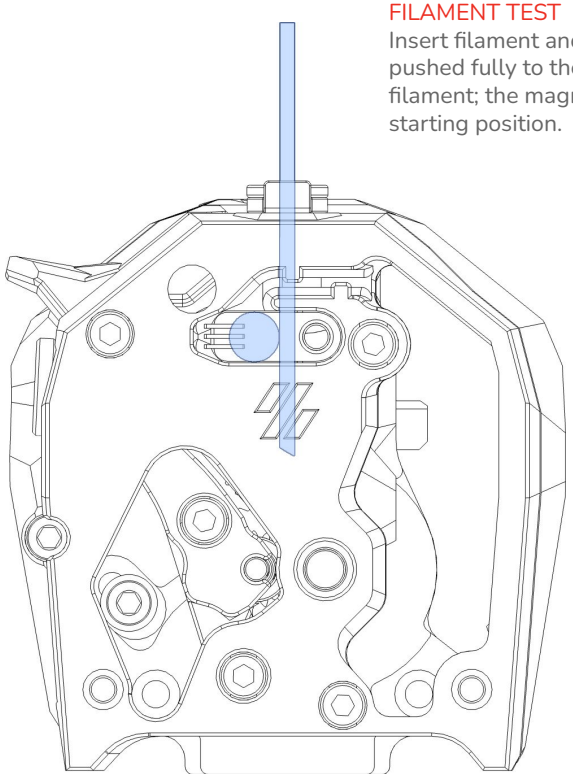
INSTALL THE SENSOR

The printed bracket and sensor will retain the magnet in the channel. Insert the assembly and secure with a ferrous/magnetic M3x25 SHCS. Do not fully tighten. The magnet must have clearance to move horizontally.



FILAMENT TEST

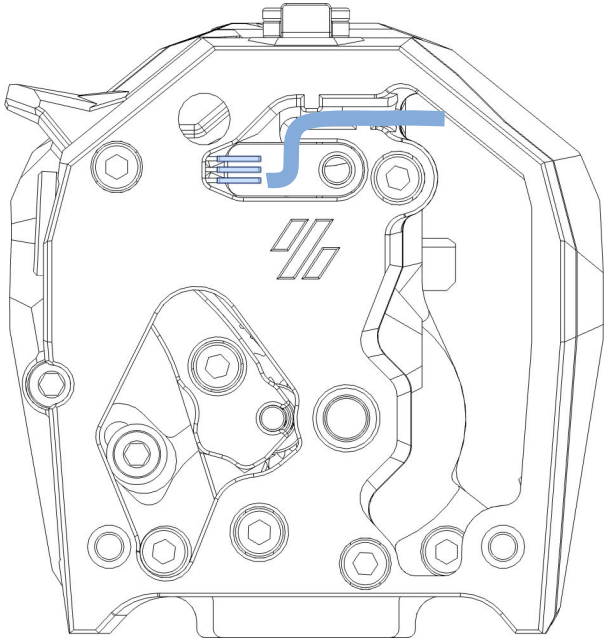
Insert filament and verify that the magnet is pushed fully to the side. Remove the filament; the magnet should return to its starting position.



HALL EFFECT SENSOR (OPTIONAL)

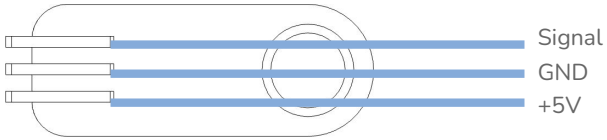
TIDY UP

Route the wires as shown.



PINOUT

Verify the sensor pinout matches the diagram when viewed from the front.



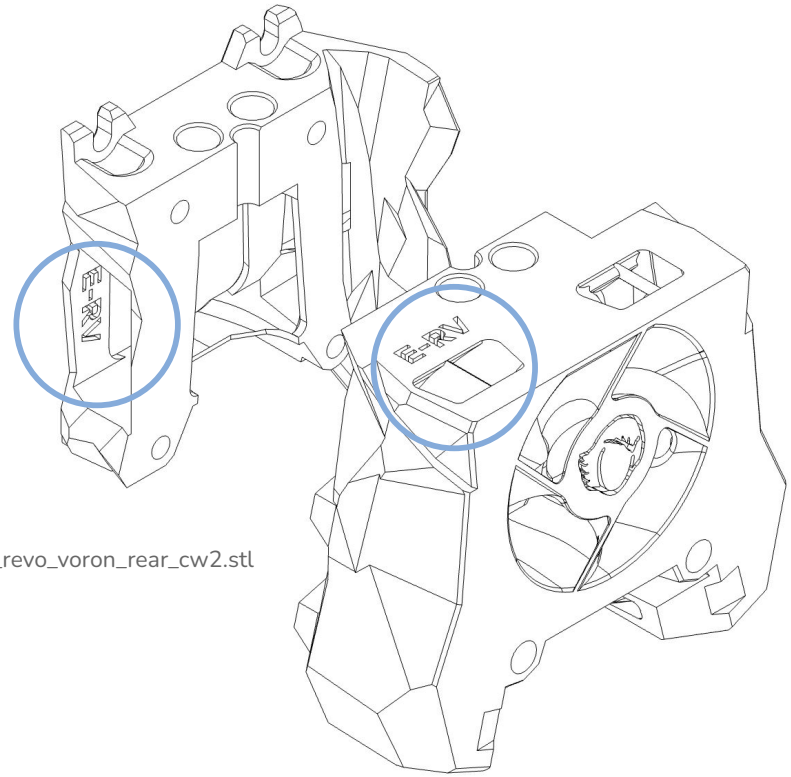
42

StealthBurner development started on 2021-04-13 with a simple: “I’m going to regret saying this but, ‘How hard could it be?’”


E-RV

IDENTIFYING PRINTED PARTS

The hotend type is embossed on the printed parts. Make sure they match your hotend.



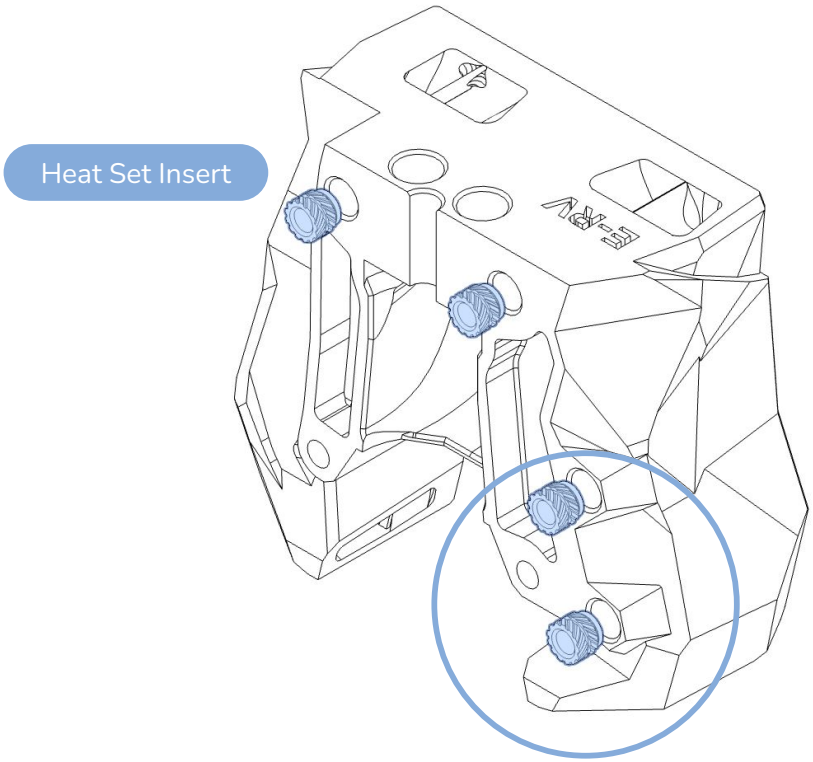
stealthburner_printhead_revo_voron_rear_cw2.stl

stealthburner_printhead_revo_voron_front.stl

AVAILABLE MOUNTS

We also provide mounts for other hotends. They are assembled in a similar manner.

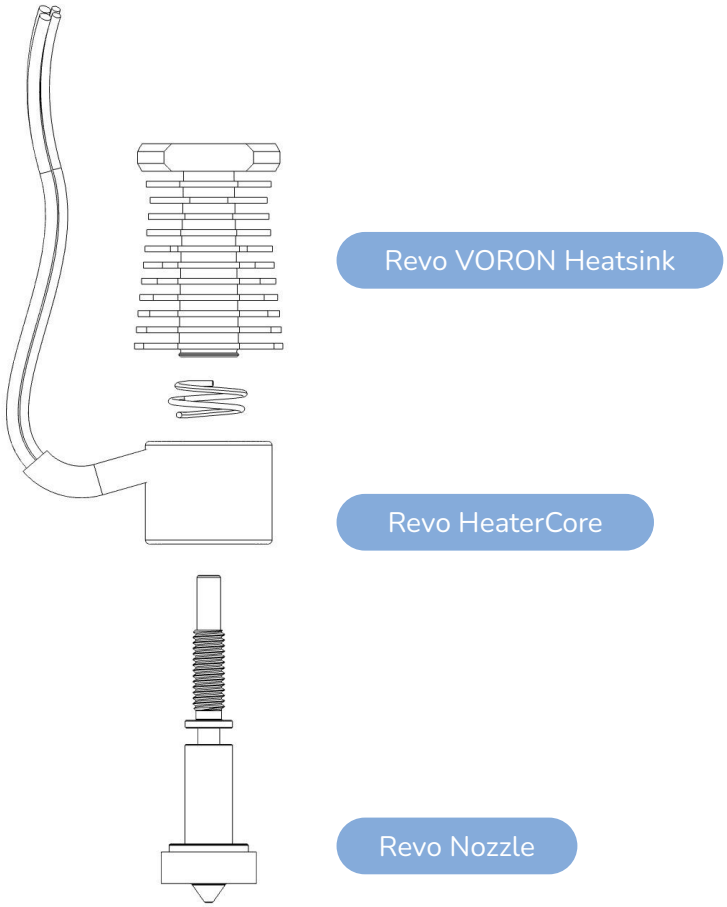
For a full list of available mounts and their identification code, see the readme file included in the toolheads folder.

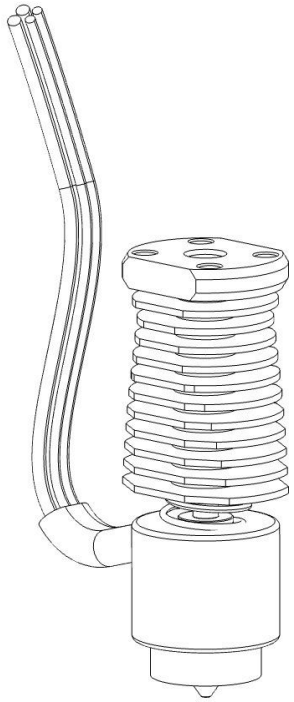


OPTIONAL: ADXL PCB MOUNT

To use the optional ADXL PCB for Input Shaper calibration, add additional heat set inserts into the holes in highlighted location.

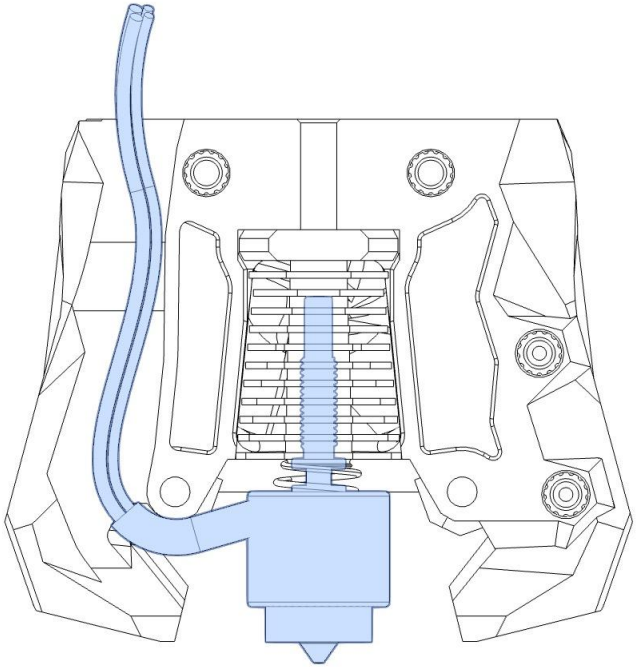
If you are using a toolhead board with an accelerometer, these do not need to be installed.


Revo VORON Heatsink
Revo HeaterCore
Revo Nozzle



HOTEND WIRE ROUTING

Route the wires as shown to the right. TI
is universal to all hotend types.



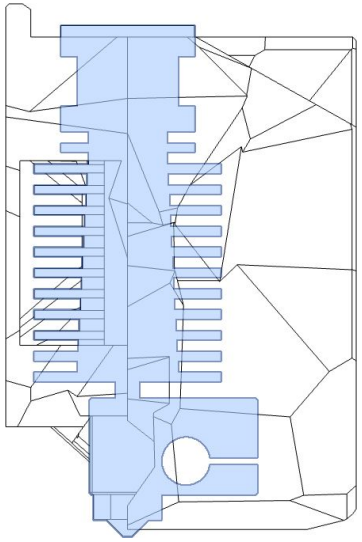
BEND STRAIN RELIEF

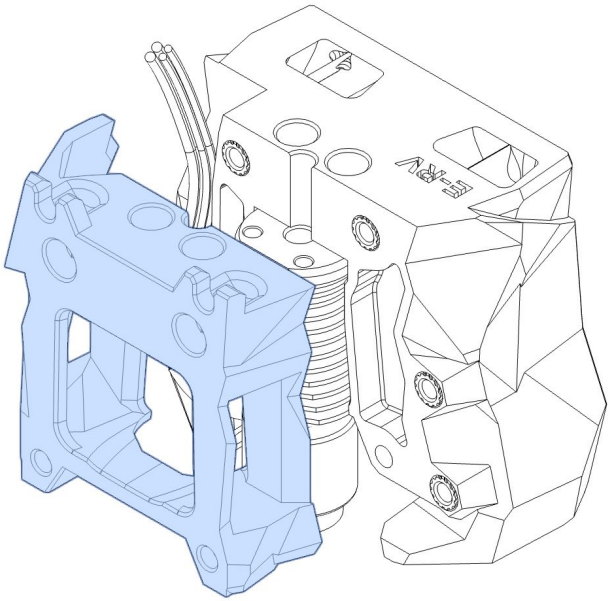
Carefully bend the strain relief to clear the printed part. Firmly hold the HeaterCore to prevent bending the Revo Nozzle.

It is critical that the Revo Nozzle is fully hand tightened into the heatsink. You MUST bend the strain relief to achieve this.

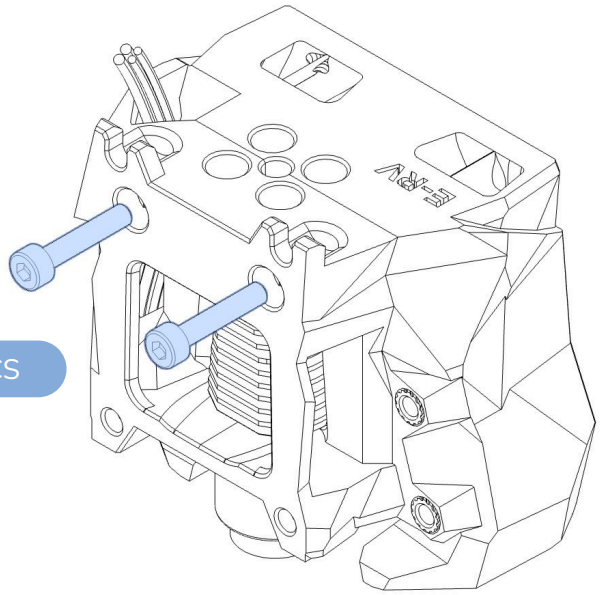
ASYMMETRICAL HEATER BLOCKS

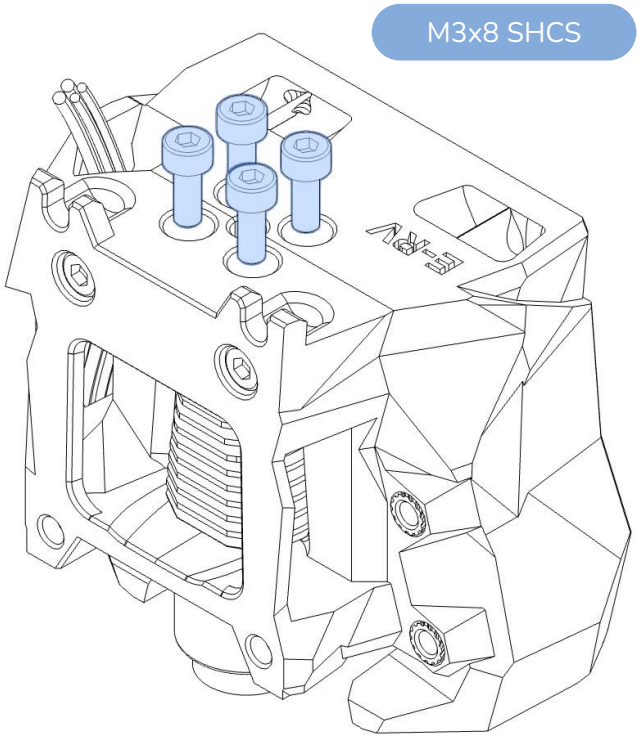
For asymmetrical heater blocks, like V6 or Dragon HF, ensure the heater block is oriented towards the front. It should not touch the rear piece.





M3x16 SHCS

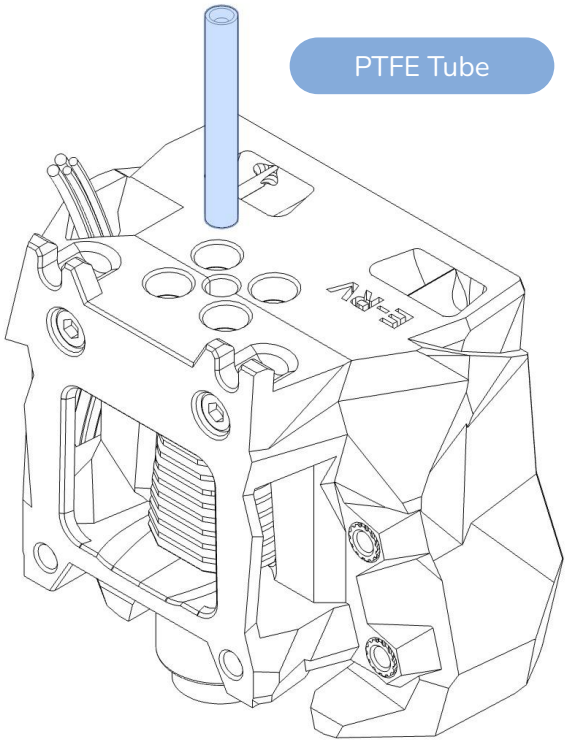




M3 MOUNTING BOLTS?

Bolts are hotend specific, Revo VORON uses M3 hardware as an improvement.

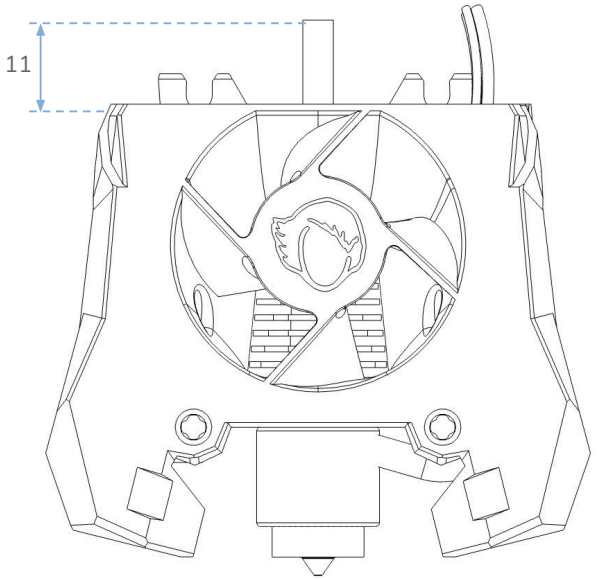
This is the way.



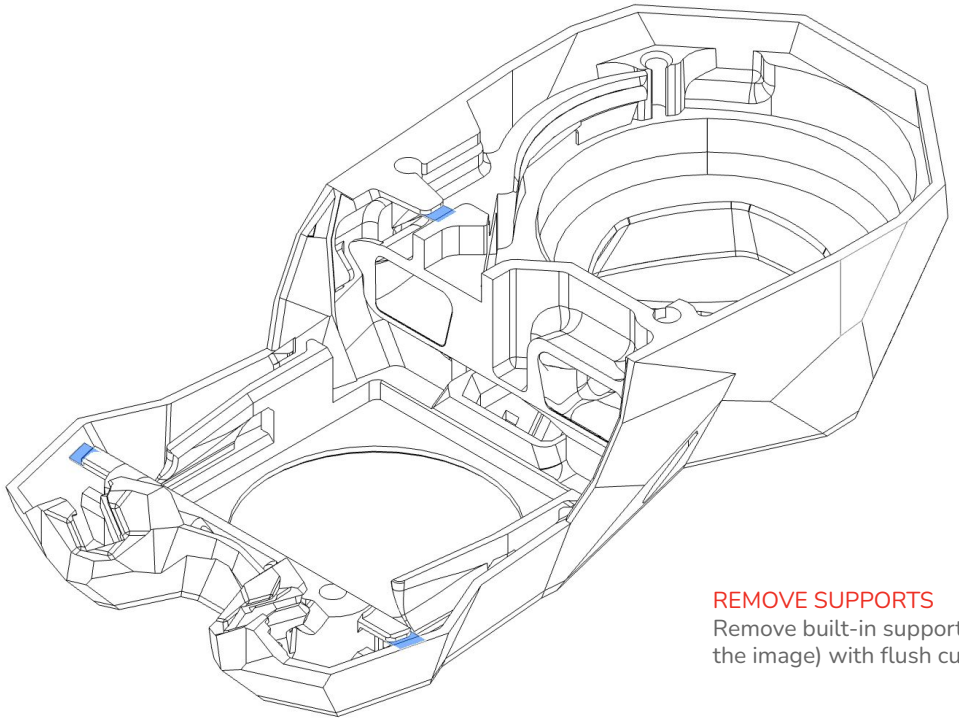
PTFE STICK OUT

The PTFE tube should stick out 11mm above the surface of the printed part.

The stick out length might vary if you use an extruder other than the Clockwork 2.5





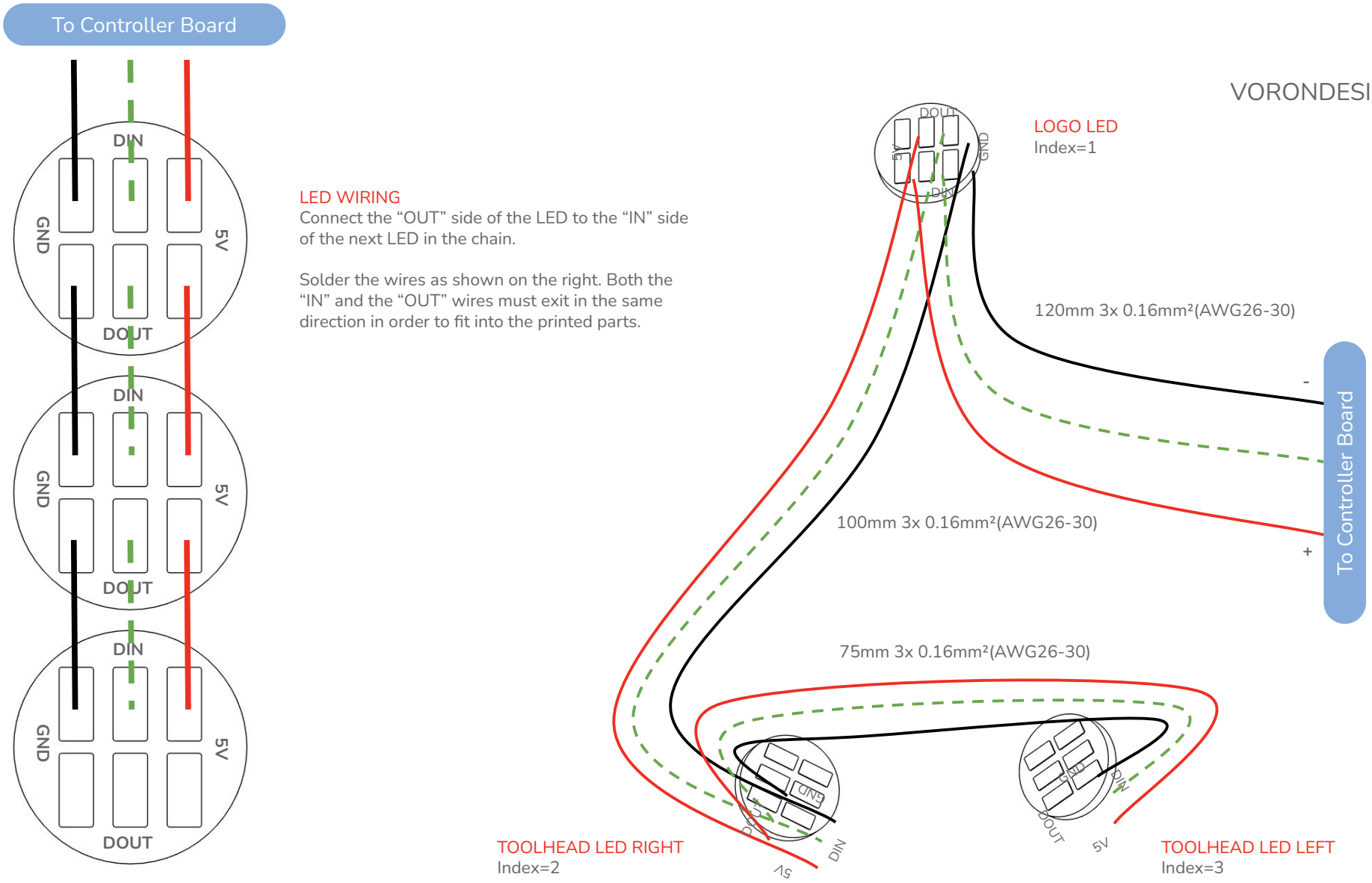


[a]_stealthburner_main_body.stl

REMOVE SUPPORTS

Remove built-in supports (highlighted in the image) with flush cutters.

For Illustration Only, Read "LED Wiring" Callout

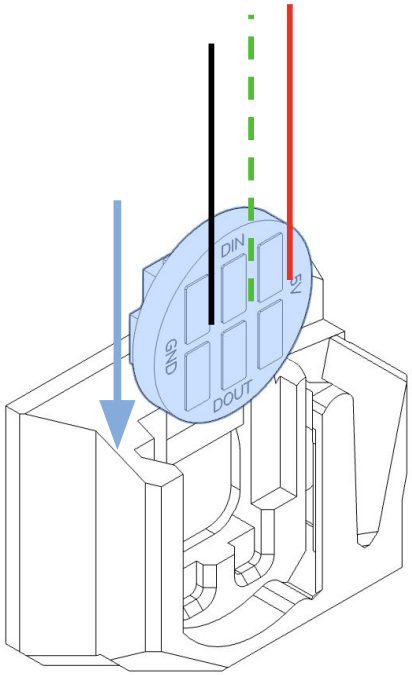


LED WIRING

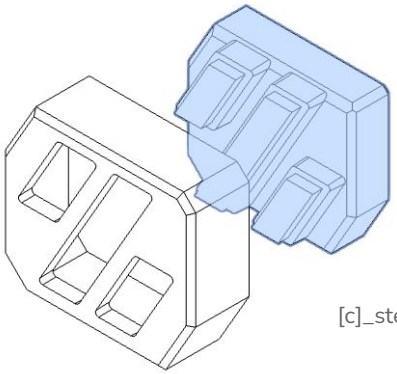
Connect the "OUT" side of the LED to the "IN" side of the next LED in the chain.

Solder the wires as shown on the right. Both the "IN" and the "OUT" wires must exit in the same direction in order to fit into the printed parts.

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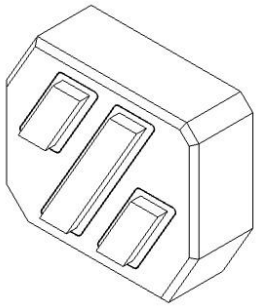


[o]_stealthburner_LED_carrier.stl



[c]_stealthburner_LED_diffuser.stl

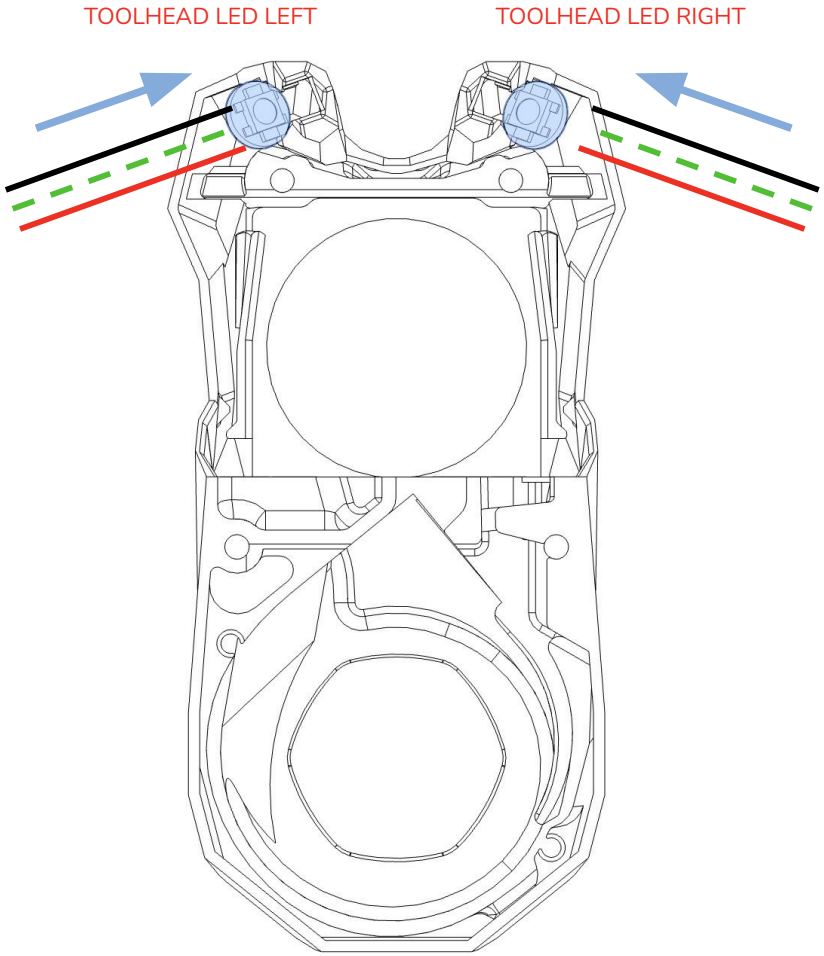
[o]_stealthburner_LED_diffuser_mask.stl

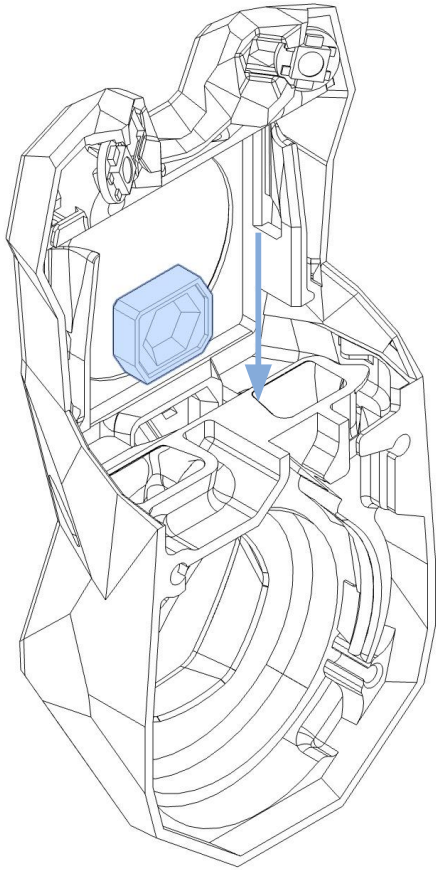


LED DIFFUSER AND MASK

The diffuser (highlighted part) is printed in a translucent filament to evenly spread the light.

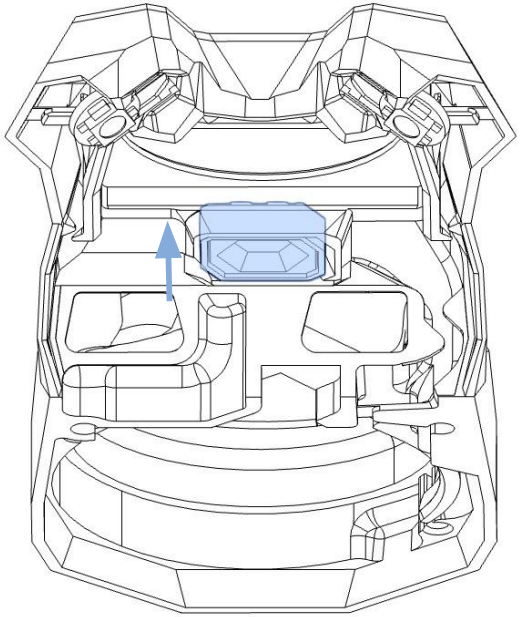
The mask (part to the left) is printed in an opaque filament to prevent the light from leaking, resulting in a crisp logo.

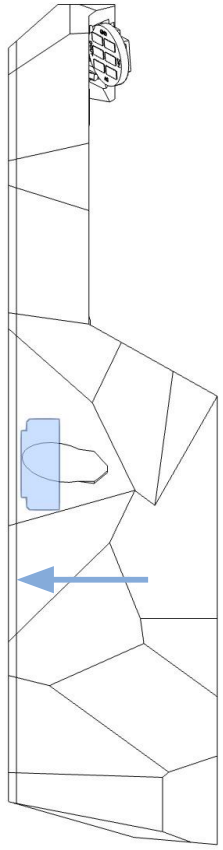

TOOLHEAD LED LEFT
TOOLHEAD LED RIGHT

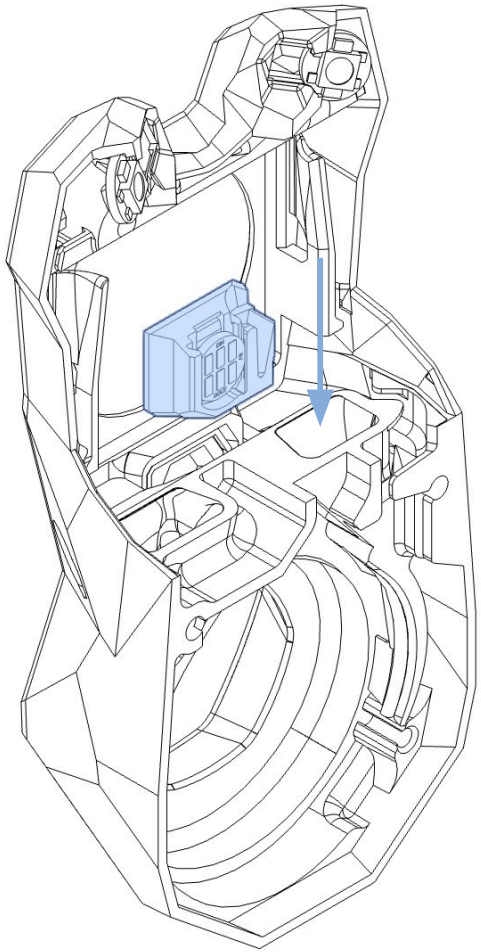


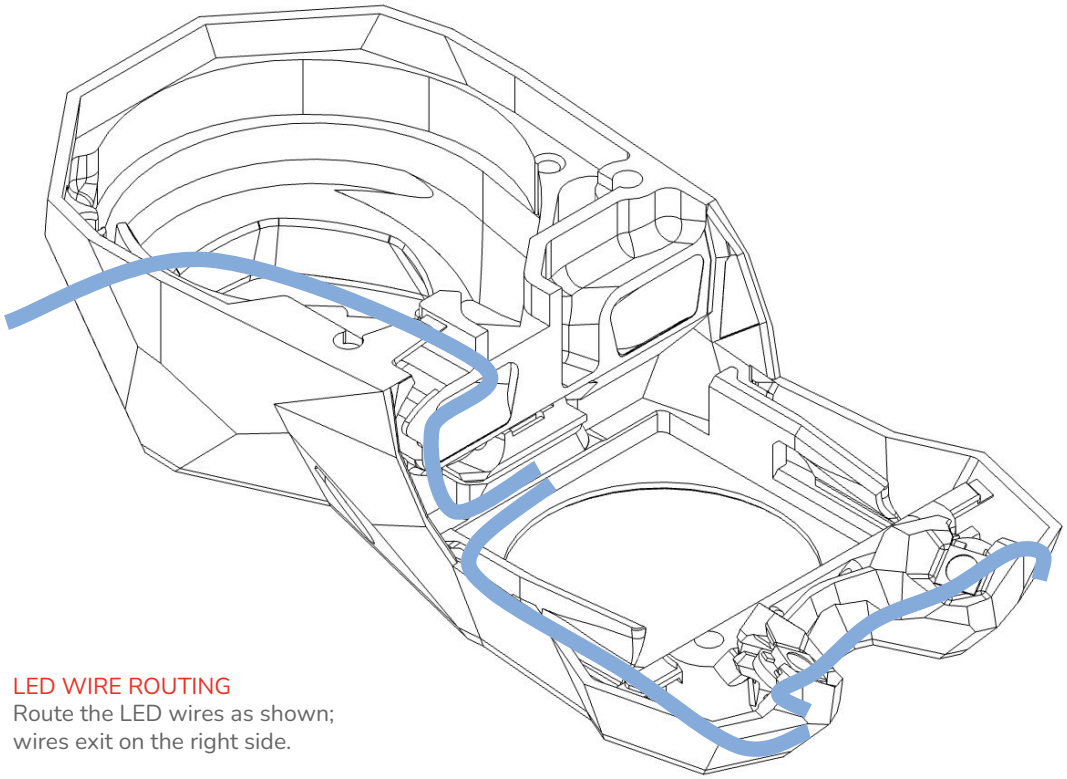
DIFFUSER INSERTION

Insert the printed parts and push them towards the front.

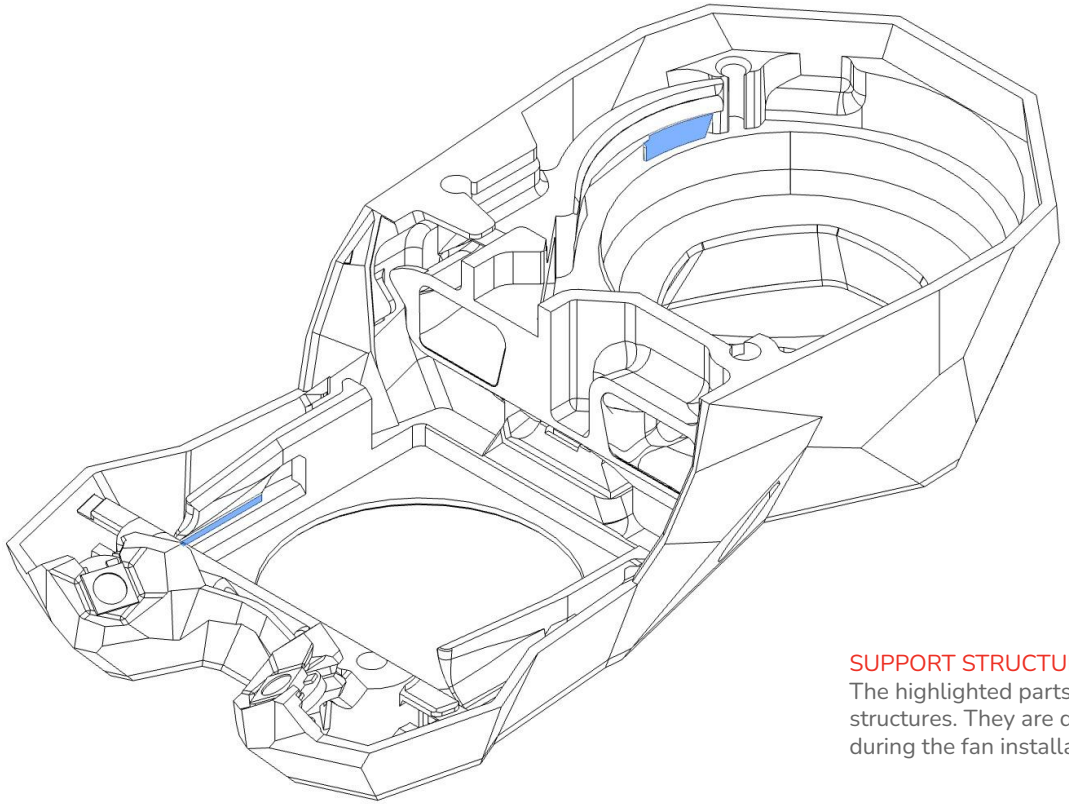






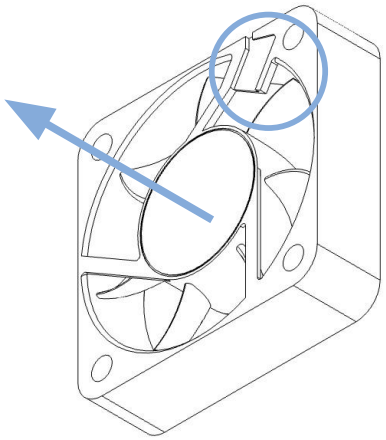


LED WIRE ROUTING
Route the LED wires as shown;
wires exit on the right side.

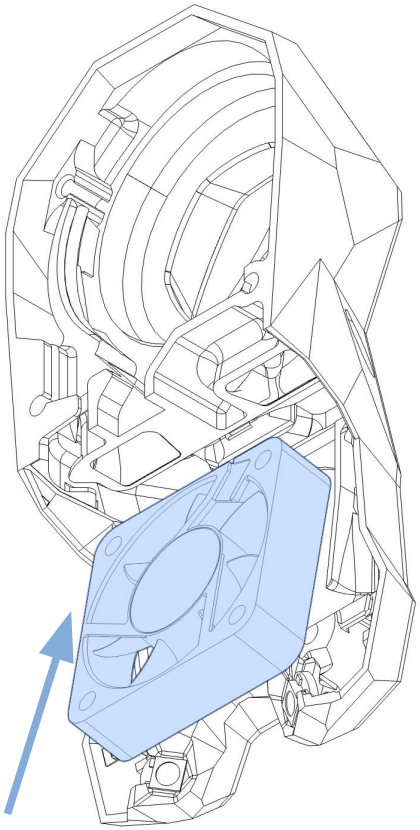


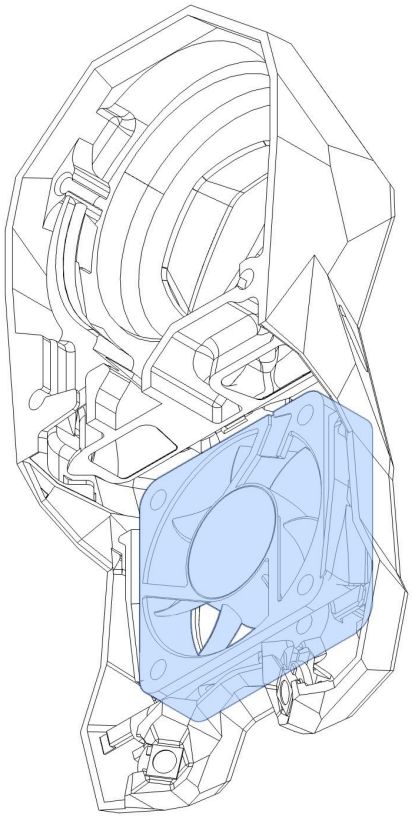
SUPPORT STRUCTURE

The highlighted parts are built-in support structures. They are designed to break during the fan installation.



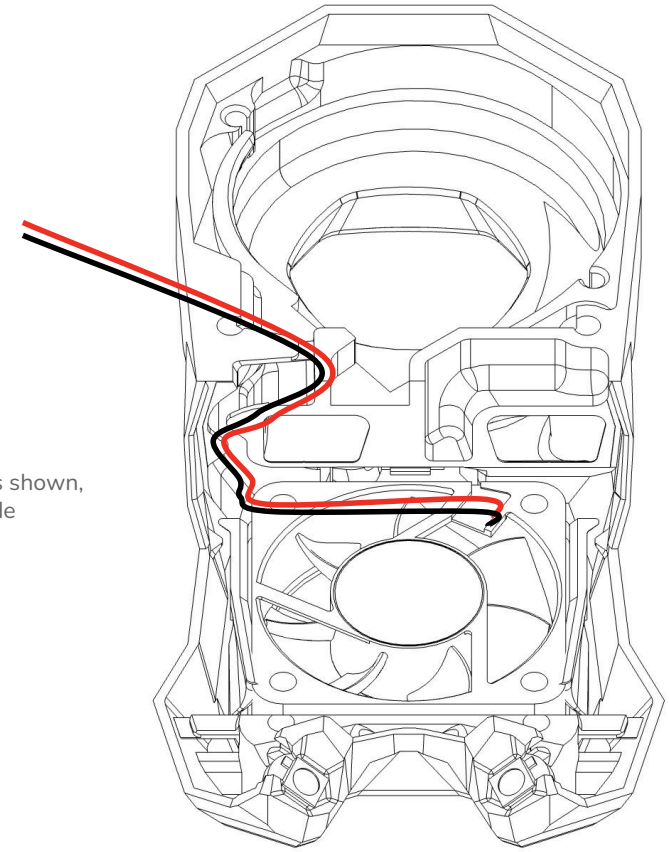
FAN ORIENTATION
Rotate the fan so that the wires exit on the top and the air is pushed “inwards”.





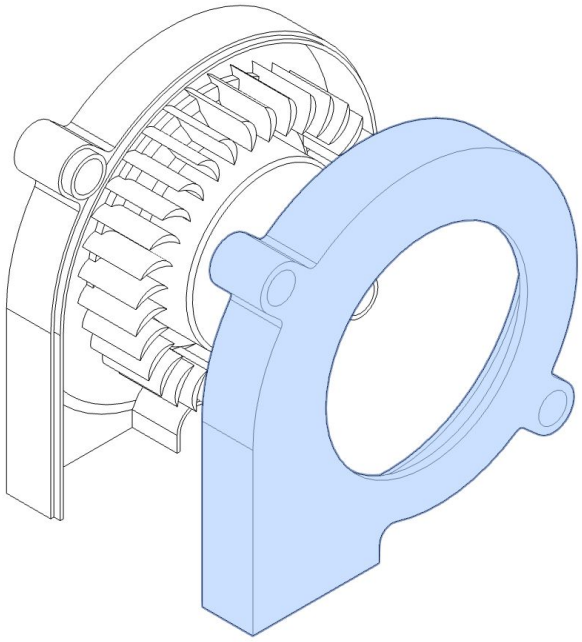
FAN INSERTION
Insert the fan at a slight angle and clip it into place.

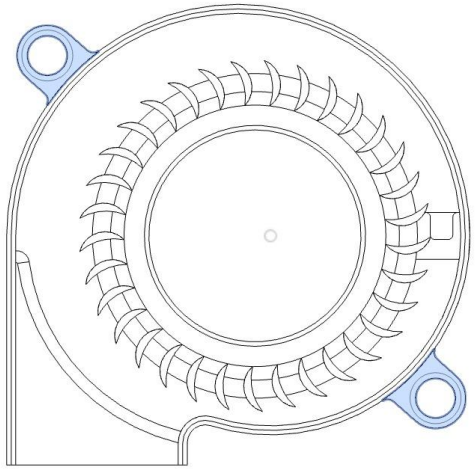
Mind the fan orientation.



FAN WIRE ROUTING

Route the fan wires as shown, exiting on the right side

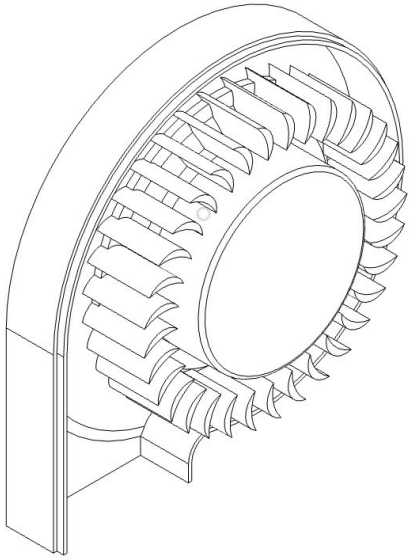


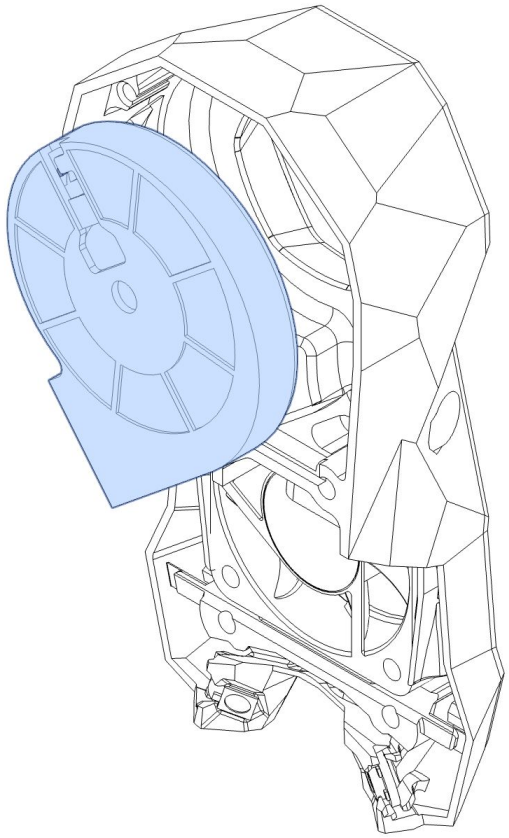


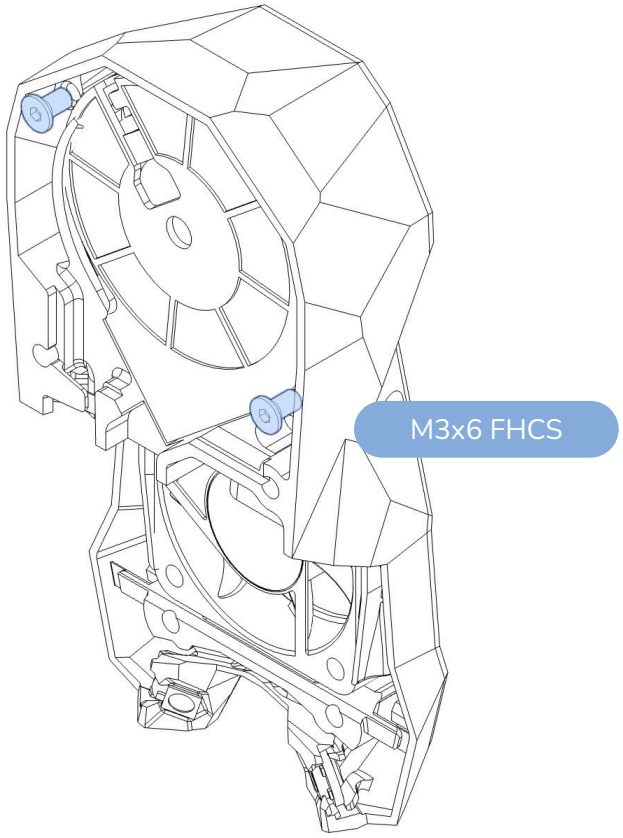
FAN PREPARATION

Remove the front of the 5015 fan. Clip off and file down the stock mounting ears.

There's a trimming jig STL included in the release to make this task easier.



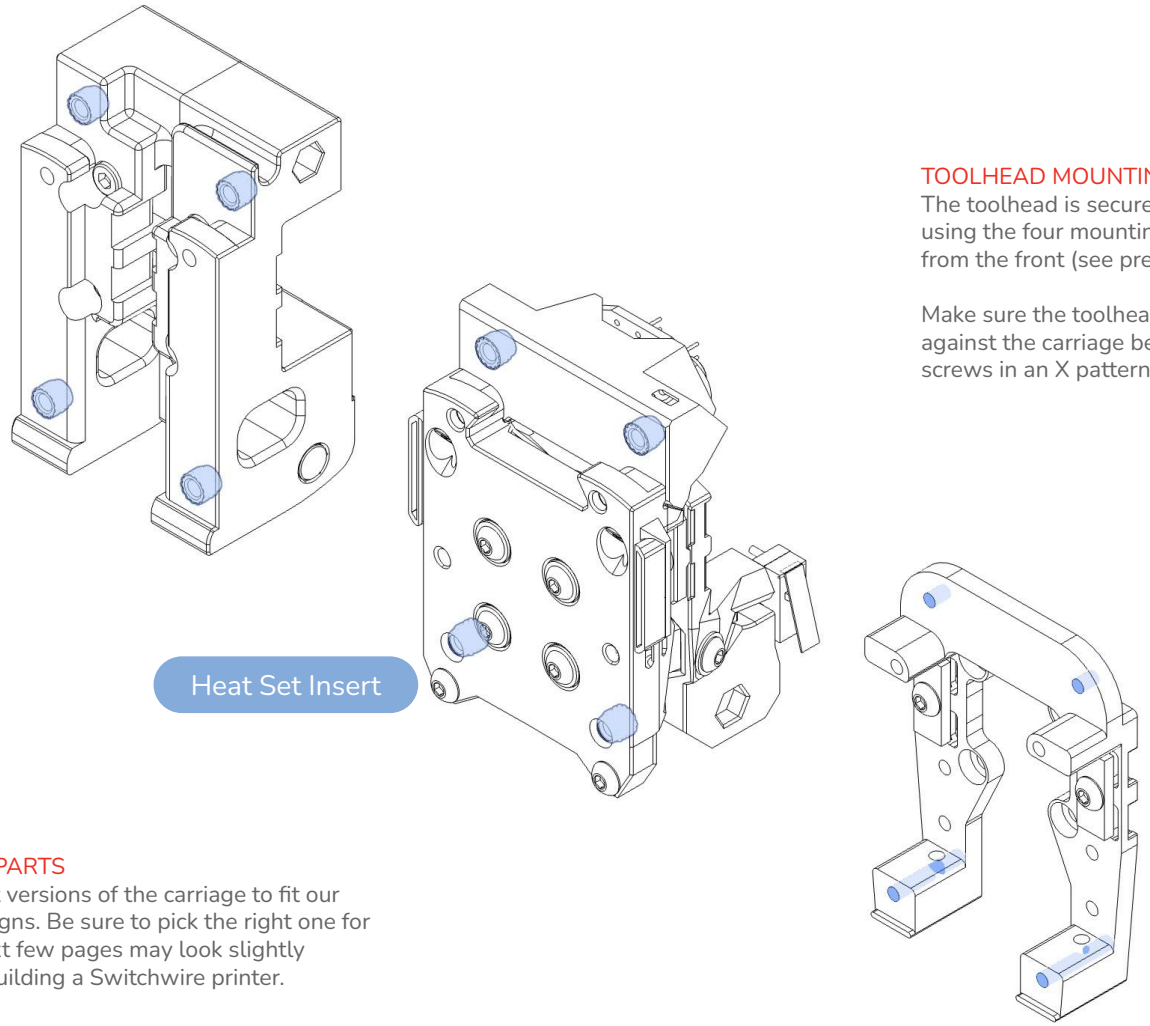



M3x6 FHCS

Yeah, it went about as expected.

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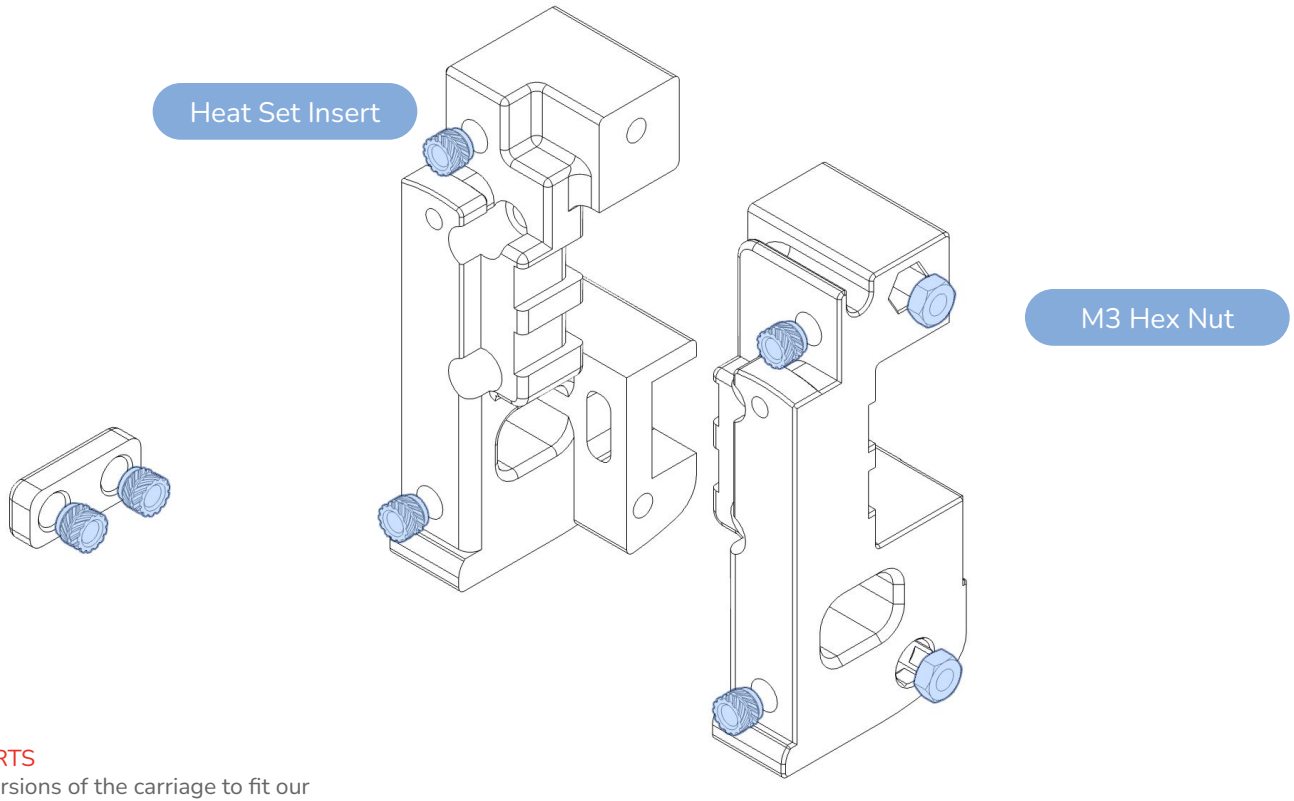
TOOLHEAD MOUNTING

The toolhead is secured to the carriage using the four mounting screws inserted from the front (see previous page).

Make sure the toolhead is seated flat against the carriage before tightening the screws in an X pattern.

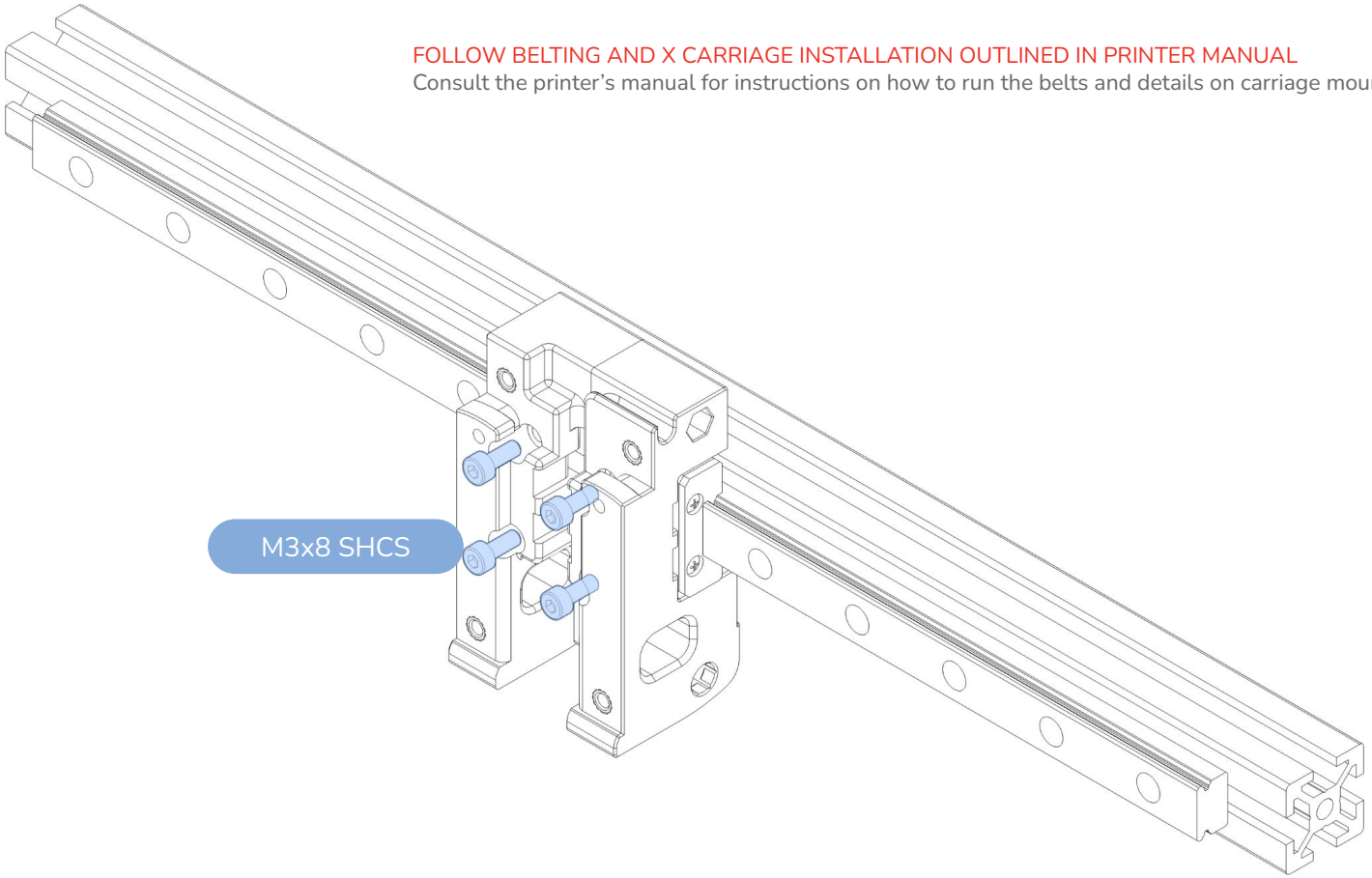
PRINTER SPECIFIC PARTS

We provide different versions of the carriage to fit our different printer designs. Be sure to pick the right one for your printer. The next few pages may look slightly different if you are building a Switchwire printer.



PRINTER SPECIFIC PARTS

We provide different versions of the carriage to fit our different printer designs. Be sure to pick the right one for your printer. The next few pages may look slightly different if you are building a Switchwire printer.

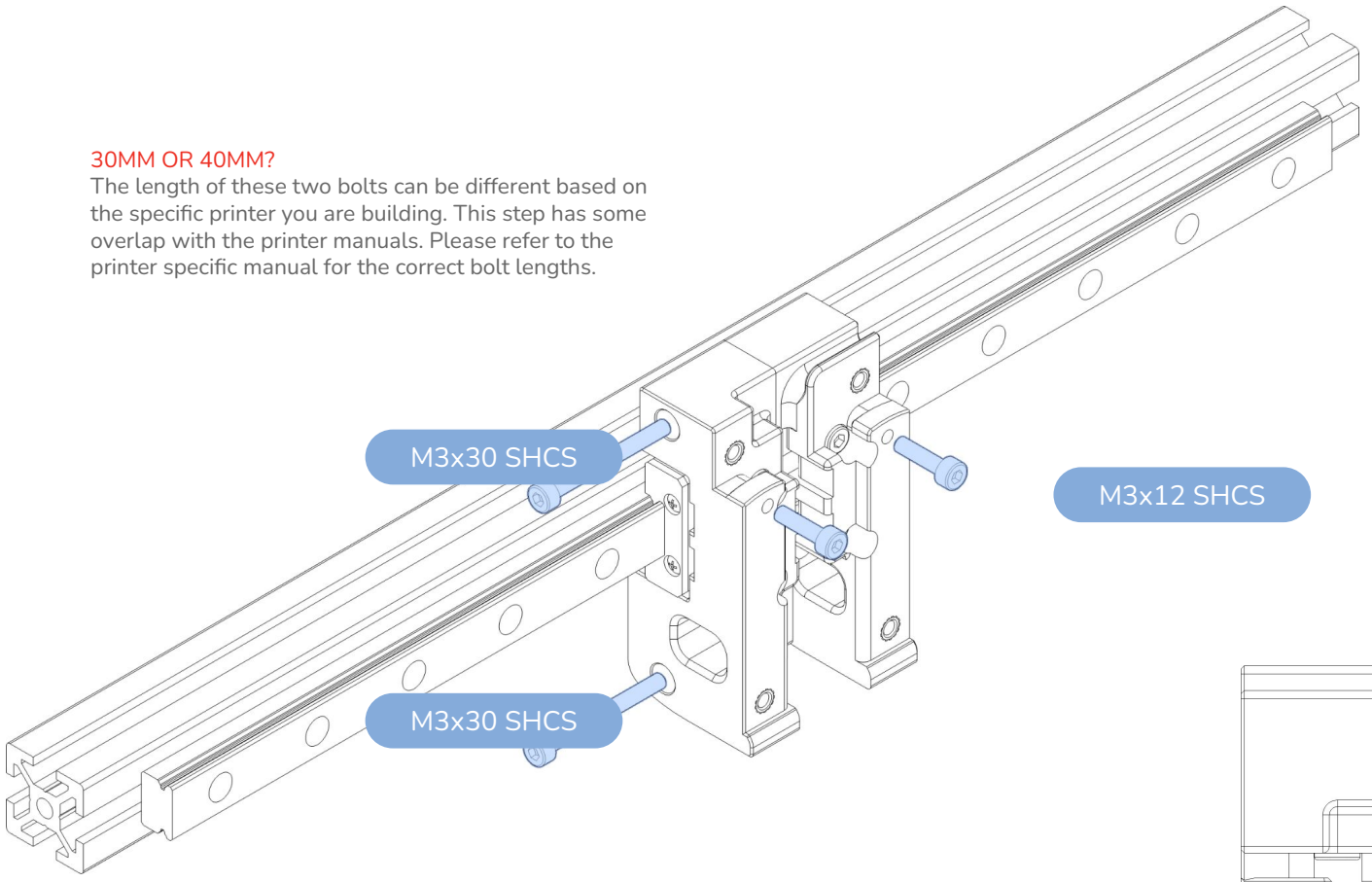


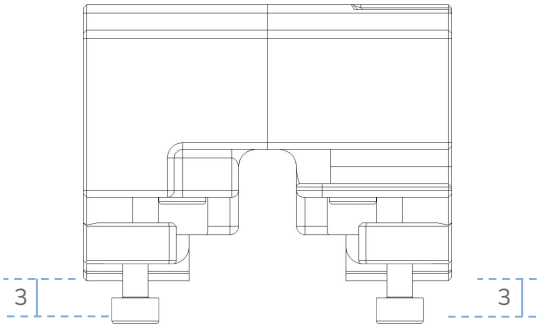
FOLLOW BELTING AND X CARRIAGE INSTALLATION OUTLINED IN PRINTER MANUAL

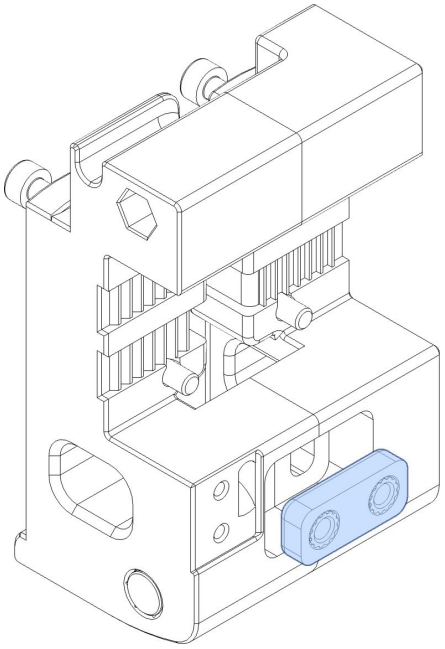
Consult the printer's manual for instructions on how to run the belts and details on carriage mounting.

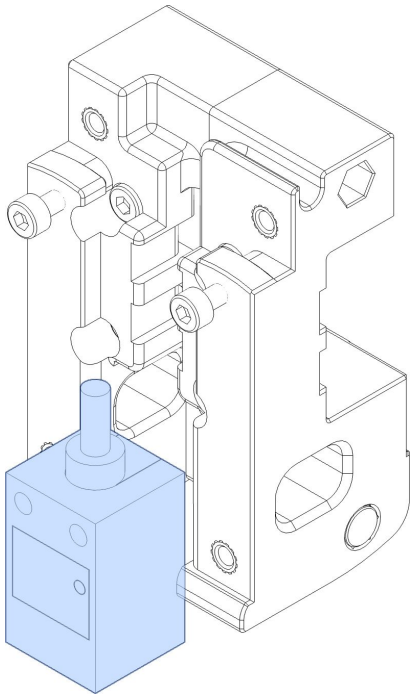
30MM OR 40MM?

The length of these two bolts can be different based on the specific printer you are building. This step has some overlap with the printer manuals. Please refer to the printer specific manual for the correct bolt lengths.







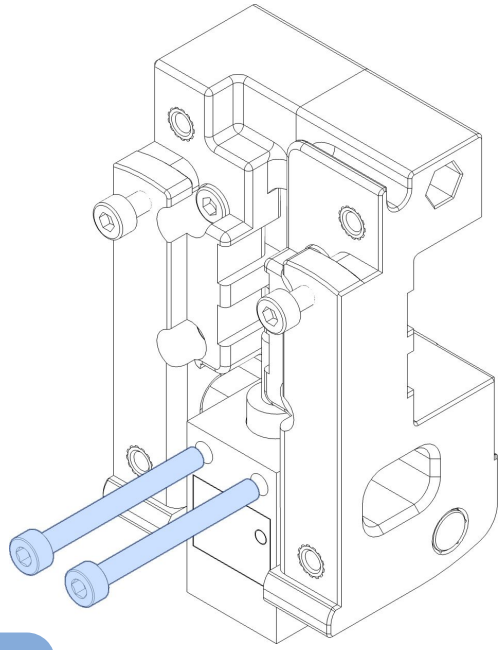


Omron Style Probe

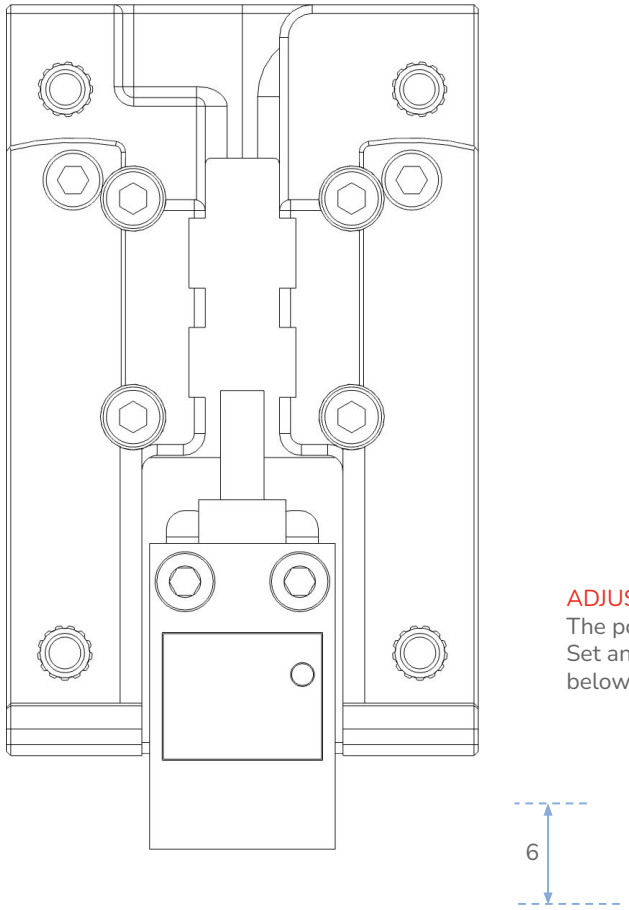
OTHER PROBE TYPES

The picture shows the BOM-specified Omron TL-Q5MC probe.

Other probes with a similar form factor and characteristics might work as well.

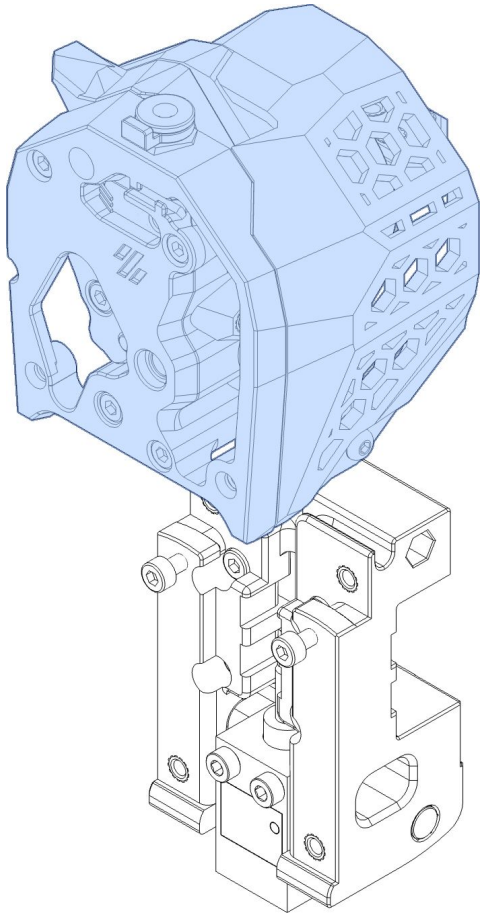


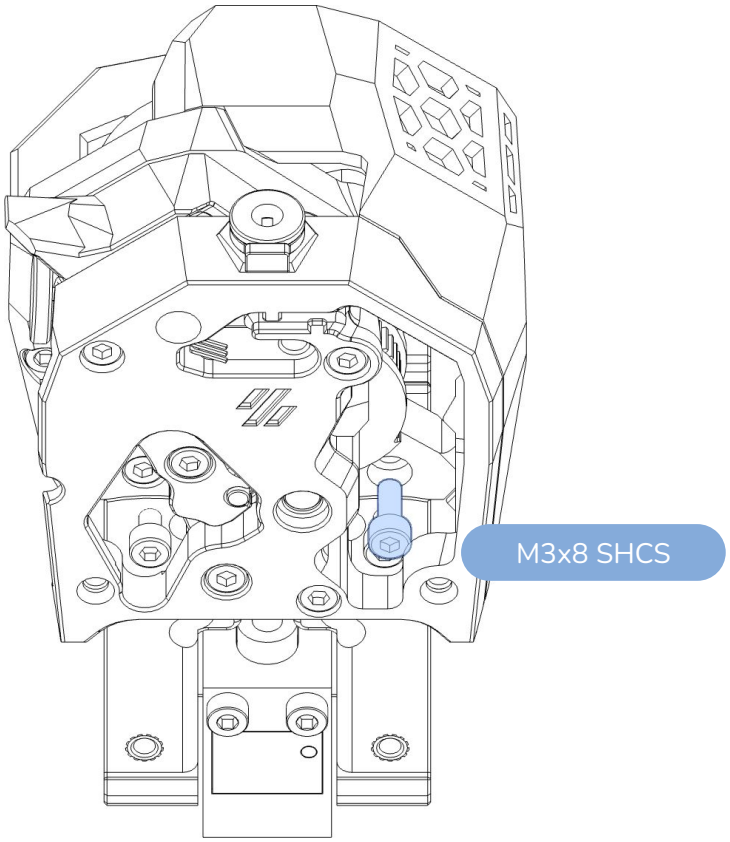
M3x30 SHCS

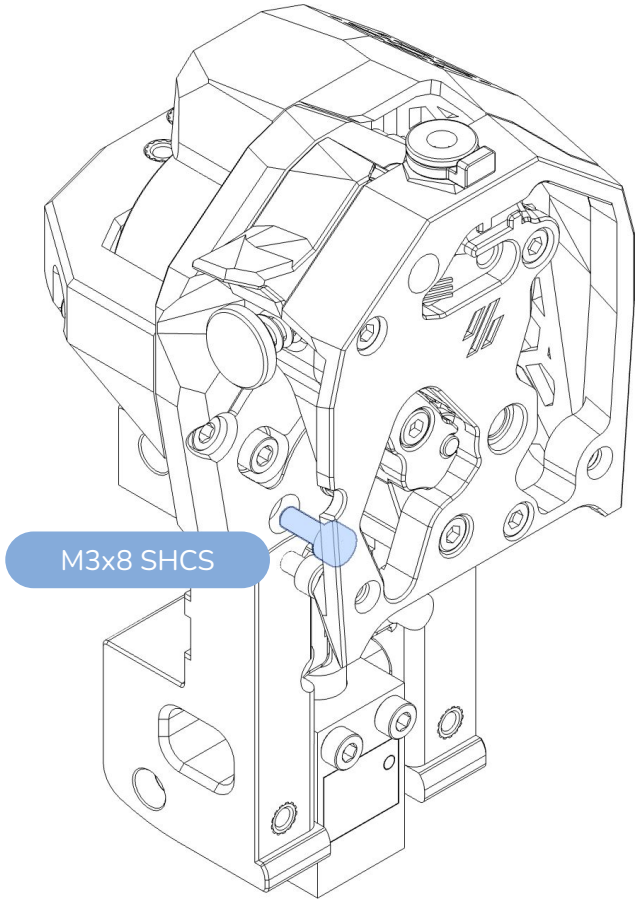


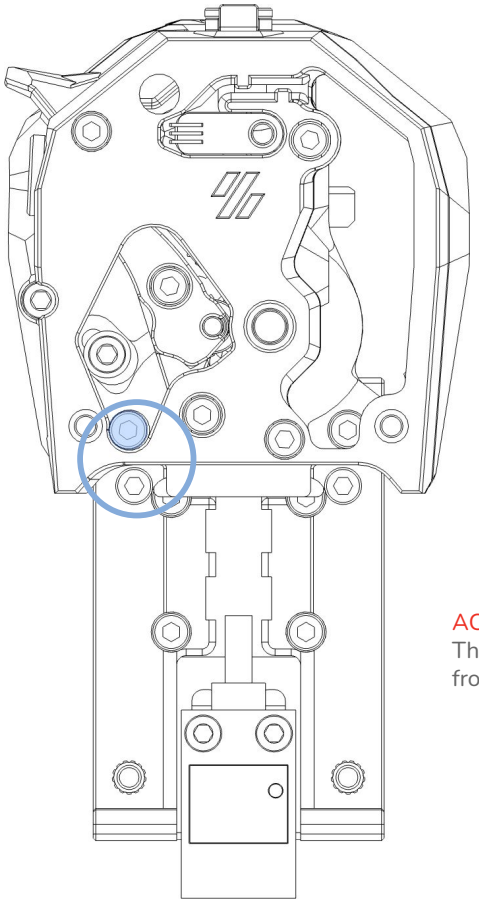
ADJUST PROBE POSITION

The position can be fine-tuned later. Set an initial position of about 6mm below the plastic part.



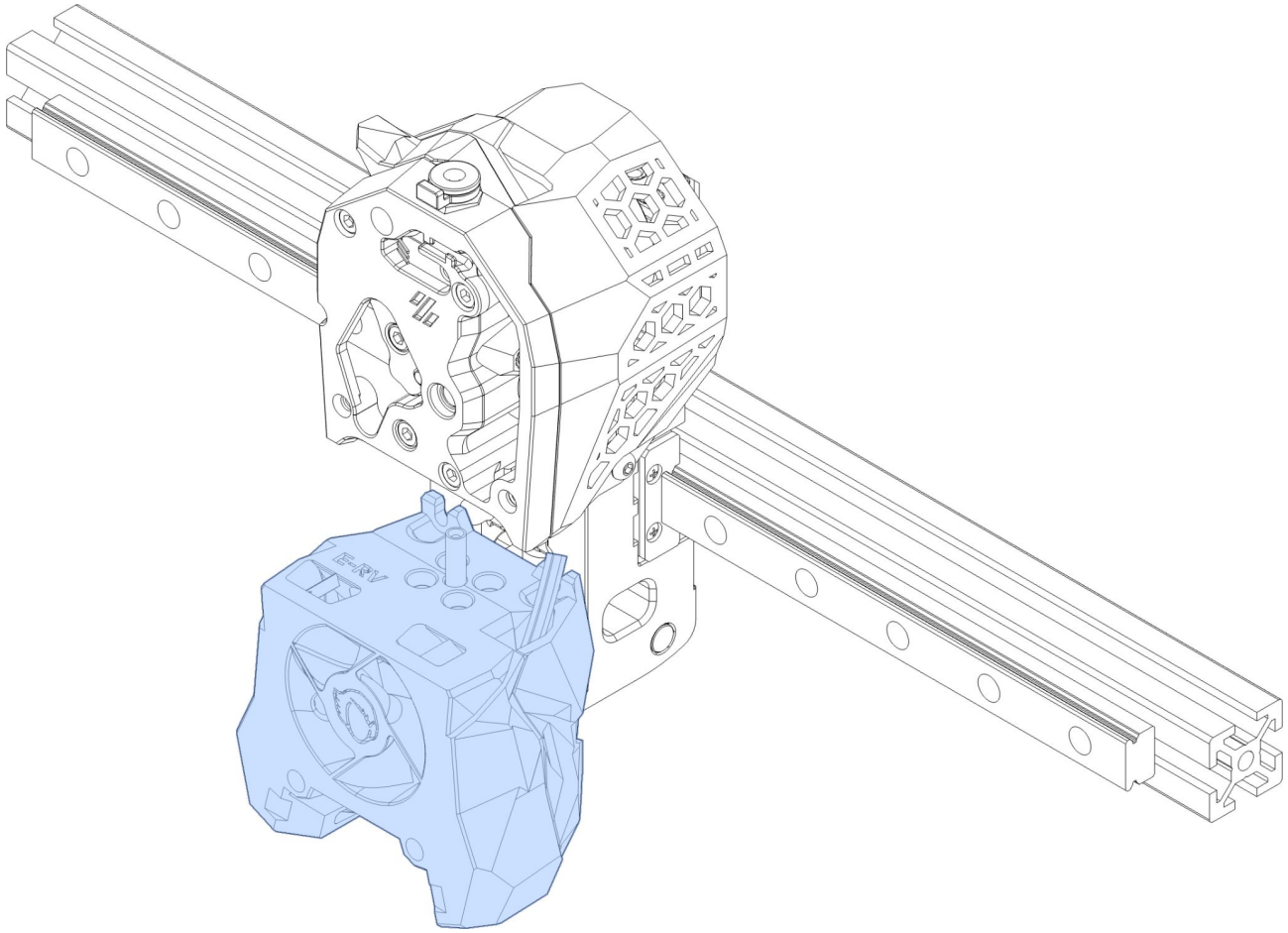

M3x8 SHCS

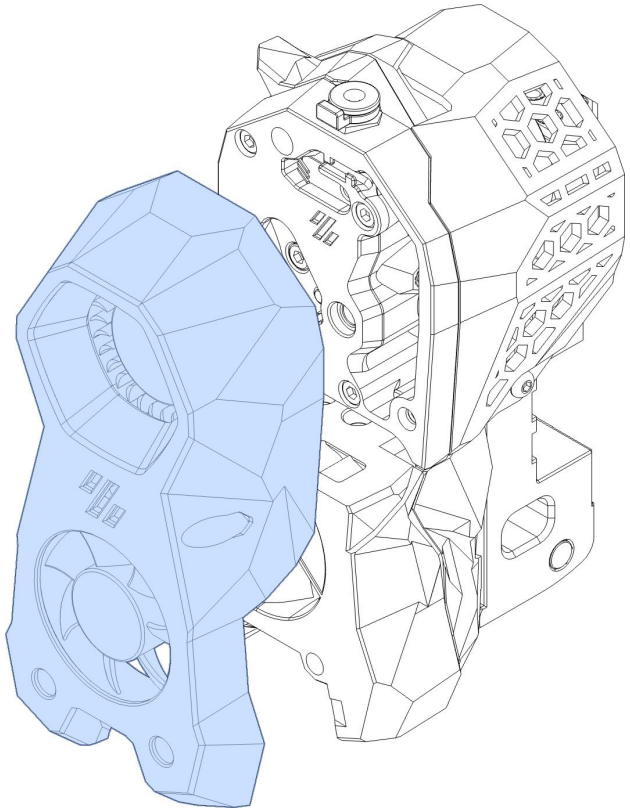


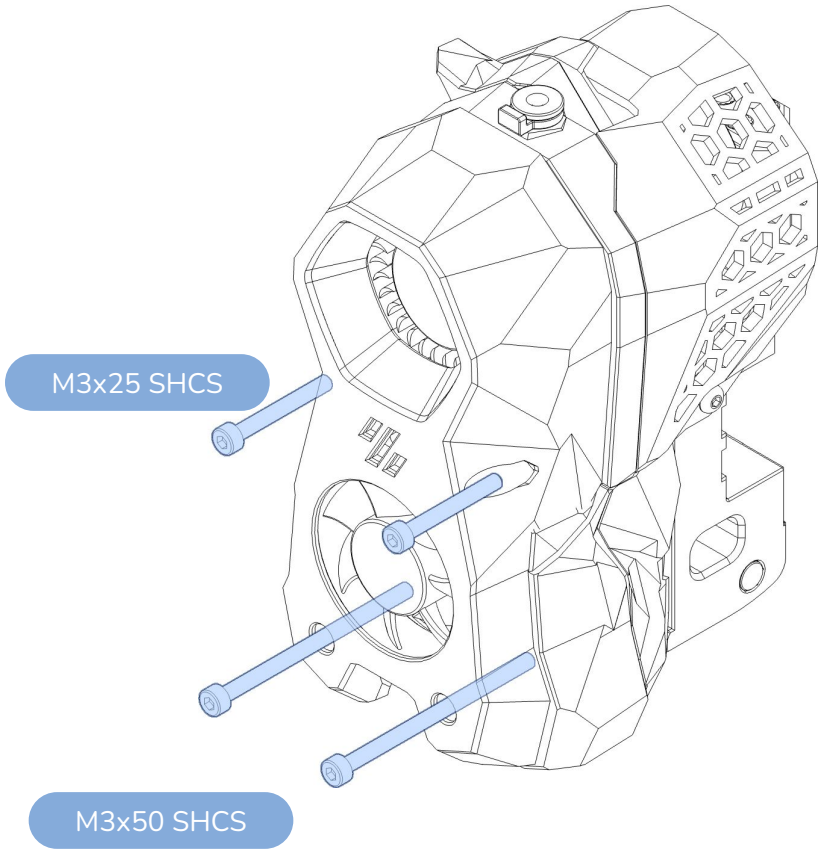


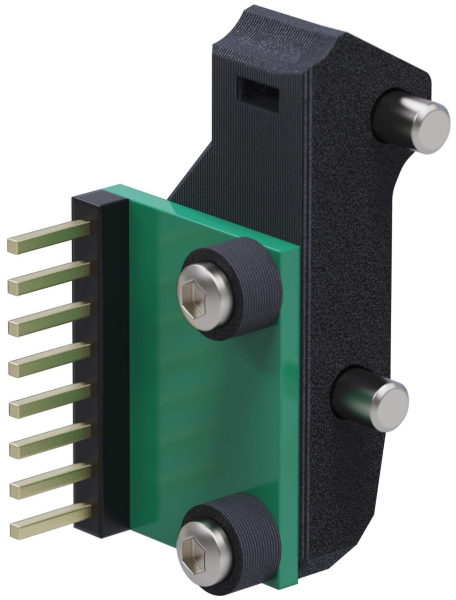
ACCESS HOLE

The bolt can be accessed from the front of the extruder.



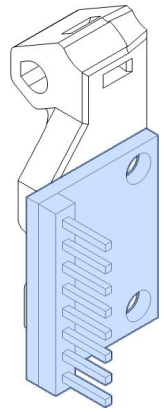



M3x25 SHCS
M3x50 SHCS



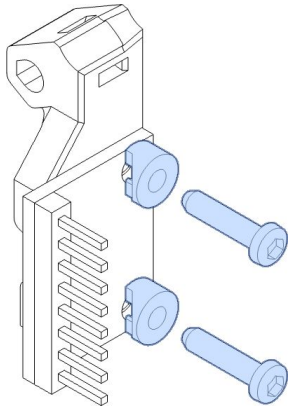
ADXL MOUNT AND INPUT SHAPER CALIBRATION

The ADXL is intended to be mounted only when doing vibration testing/input shaper calibration and should not be left on the tool head during normal printing.



sb_adxl_mount_generic_15.5mm_c_c.stl

ADXL345 Prototype Board

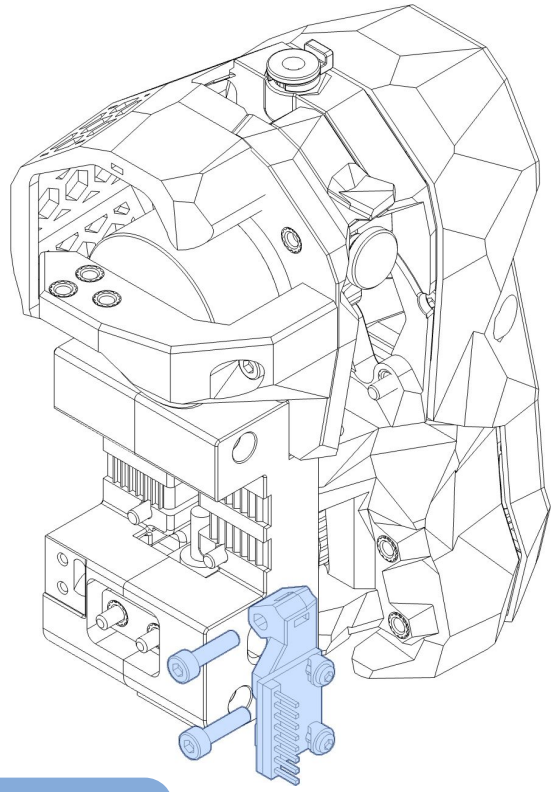


sb_adxl_washer_x2.stl

M2x10 Self Tapping Screws

ISOLATION SPACERS

Most ADXL345 prototype boards have circuit tracks close to the mounting holes. Make sure to install spacers to prevent damage to the board. We included printable spacers in the release.



M3x12 SHCS

MOUNTING TO THE TOOLHEAD

You can use M3x12 screws to mount this assembly to your toolhead.

ASSEMBLY COMPLETED! ... NEXT STEP: SETUP & CALIBRATION

This manual is designed to be a reference manual for the build process of a VORON Stealthburner Toolhead. Additional details about the build and background on advanced topics can be found on our documentation page linked below.

The software setup and other initial setup steps with your new toolhead can also be found on our documentation page. We recommend starting here.



https://docs.vorondesign.com/



https://github.com/VoronDesign/Voron-Stealthburner

HOW TO GET HELP

If you need assistance with your build, we're here to help. Head on over to our Discord group and post your questions. This is our primary medium to help VORON Users and we have a great community that can help you out if you get stuck. Alternatively, you can use our subreddit.



https://discord.gg/voron



https://www.reddit.com/r/VORONDesign

REPORTING ISSUES

Should you find an issue in this document or have a suggestion for an improvement please consider opening an issue on GitHub (https://github.com/VoronDesign/Voron-Trident/issues).

When raising an issue please include the relevant page numbers and a short description; annotated screenshots are also very welcome.

We periodically update the manual based on the feedback we get.

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Website
vorondesign.com

GitHub
github.com/vorondesign

Docs
docs.vorondesign.com

Discord
discord.gg/voron

